



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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https://github.com/CarlosJunior41/Space_Y_project



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

- Data collection from api.spacexdata.com/v4/launches/past
- Data wrangling from https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches
- Exploratory data analysis (EDA) using visualization and SQL
- Interactive visual analytics using Folium and Plotly Dash
- Predictive analysis using classification models to compare results

Summary of all results

- The comparison of the models using `best_score_` from cross-validation shows that Decision Tree had the highest accuracy (85.7%).
- Although different algorithms were tested, all produced identical results on the test set, suggesting that the problem may be relatively simple or that the size of the dataset limits the ability to distinguish performance between models.
- The recall for the Did not Land class is relatively low (0.50), indicating that the model fails to correctly identify half of the cases in which the rocket does not land. Predicting that a rocket will land when it will not can be very costly.

For this application, this could represent a significant risk and suggests the need for improvements to the model or data balancing.

Introduction

- Context
- Business question

The commercial space age is here, companies are making space travel affordable for everyone

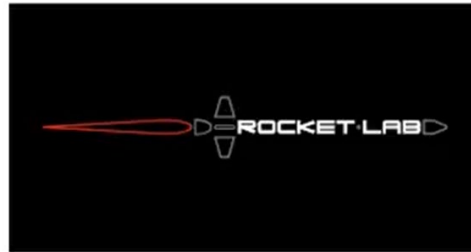
Virgin Galactic is providing suborbital spaceflights



Revenue of \$7 million for each of the years ended December 31, 2024 and 2023, which was primarily attributable to commercial spaceflights and memberships fees related to future astronauts.

<https://press.virgingalactic.com/virgin-galactic-announces-fourth-quarter-and-full-year-2024-financial>

Rocket Lab is a small satellite provider.



Rocket Lab Announces Fourth Quarter and Full Year 2025 Financial Results, Posts Record Quarterly Revenue of \$180M, Record Annual Revenue of \$602M, Delivering Annual Growth of 38% and Growing Backlog 73% Year-on-Year to \$1.85B

<https://investors.rocketlabcorp.com/news-releases/news-release-details/rocket-lab-announces-fourth-quarter-and-full-year-2025-financial>

Blue Origin manufactures sub-orbital and orbital reusable rockets.

BLUE ORIGIN



Blue Origin's revenue in 2025 is driven by increased suborbital flights and major government contracts, including a \$2.3 billion NSSL Phase 3 award and a \$78.2 million satellite infrastructure contract.

<https://aeromorning.com/en/growing-pressure-on-spacex-blue-origins-new-glenn-gains-momentum/#:~:text=Economic%20Stakes%20and%20Contracts,is%20a%20capital%20intensive%20endeavour.>

Commercial Space Travel 2025 Annual Review: Space travel passenger, revenue, volume, and demographics:

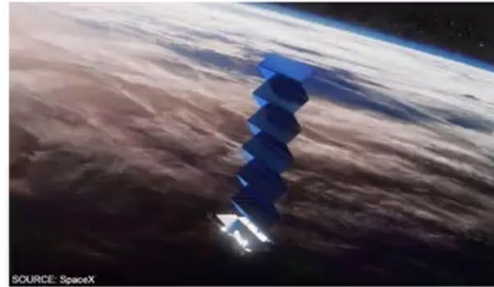
<https://www.spaceportlounge.com/p/2025-space-travel-passenger-revenue-volume-demographics>

In this new market, SpaceX has been successful



www.flickr.com

Sending spacecraft to the
International Space Station.



Source Space X

Starlink, a satellite internet
constellation providing satellite
Internet access. Sending manned
missions to Space



Source NASA

Sending manned missions to
Space

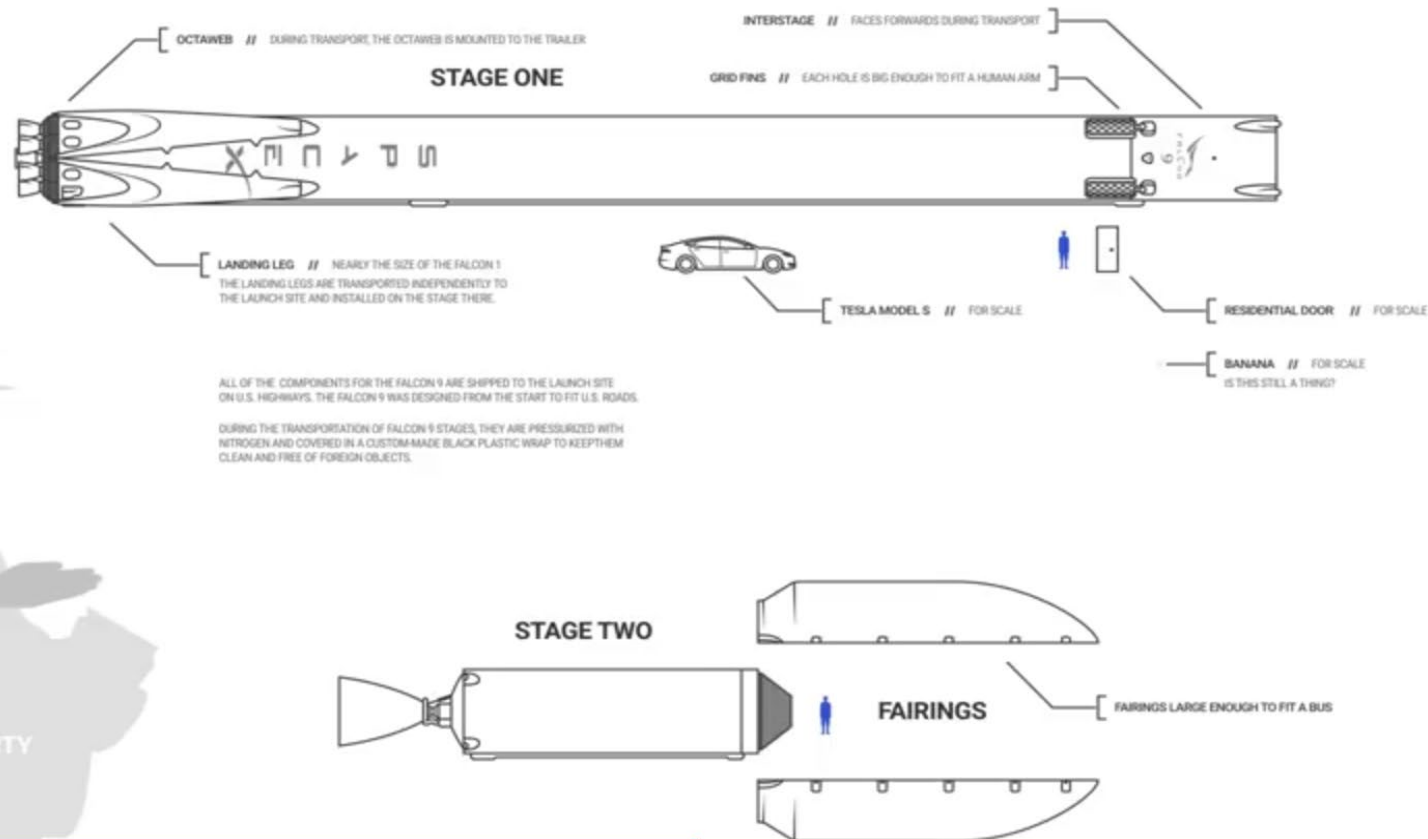
SpaceX reportedly generated a revenue of close to \$16 billion in 2025, with the company also generating an \$8 billion profit, most of which comes from its Starlink business, which accounted for 50% to 80% of the revenue.

<https://finance.yahoo.com/news/elon-musk-says-nasa-account-233113985.html>

In the rocket launch business, SpaceX has a powerful cost advantage because of the reuse of the first stage

- SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars;
- other providers cost upwards of 165 million dollars each,

SpaceX Falcon 9 // SCALE AND TRANSPORTATION
* AS OF JUNE 2016



Question:

What is SpaceX's expectation regarding the reuse of the first stage?



The objective of this work is to determine the price of each launch.

To this end, information about SpaceX will be collected and dashboards will be created to support the analysis.

This presentation shows the information gathered and its processing method for the modeling work:

- The determination of whether SpaceX will reuse the first stage will be obtained by training a machine learning model using public information to predict whether SpaceX will reuse the first stage.

Section 1

Methodology



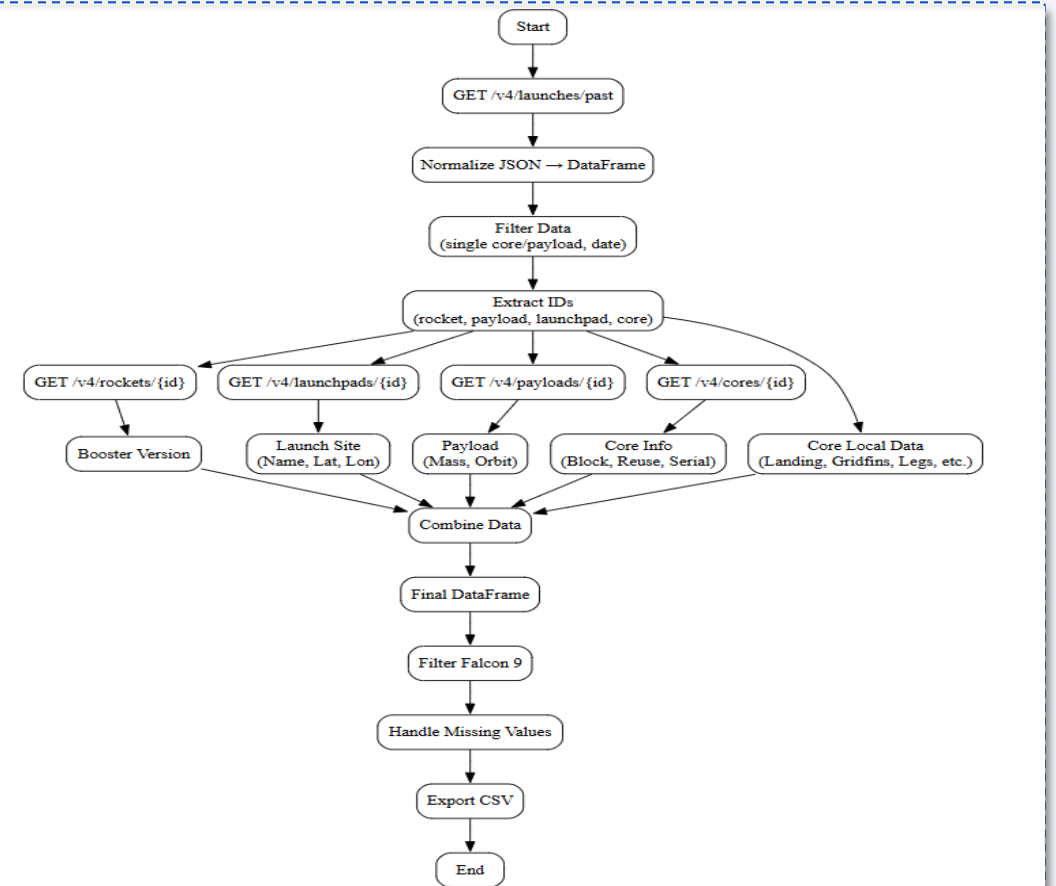
Methodology

Executive Summary

- Data collection methodology:
 - api.spacexdata.com/v4/launches/past
- Perform data wrangling
 - https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models

Data Collection – SpaceX API

- SpaceX REST API as a data source for analysis
- The received data underwent an initial cleaning process
- Selected columns, along with these, formed the initial attribute base for predicting success or failure
- To better focus the analysis, only Falcon 9 launches were selected



by <https://dreampuf.github.io/>

SpaceX REST API as a data source for analysis

Our goal is to use the SpaceX REST API data to predict whether SpaceX will attempt to land a rocket or not.

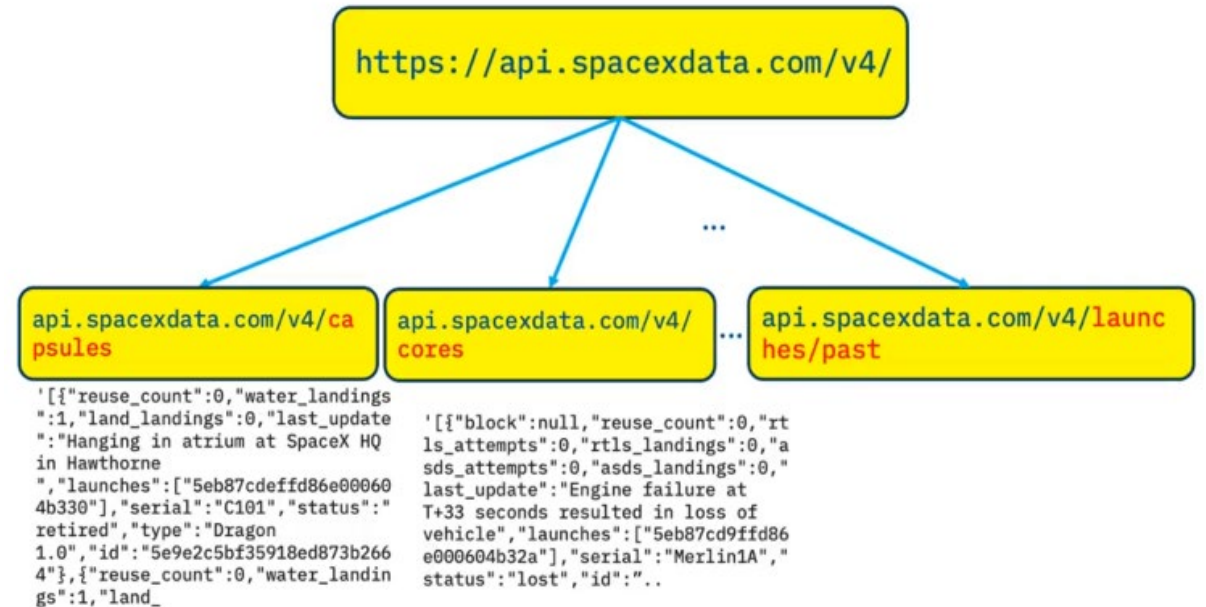
The SpaceX REST API endpoints, or URL, starts with **api.spacexdata.com/v4/**.

We have the different end points, for example: /capsules and /cores We will be working with the endpoint **api.spacexdata.com/v4/launches/past**.

This result can be viewed by calling the `.json()` method.

Specifically, we have a list of JSON objects which each represent a launch.

To convert this JSON to a dataframe, we can use the `json_normalize` function.



```
url="https://api.spacexdata.com/v4/launches/past"
```

```
response =requests.get(url)
```

```
response.json()
```


The received data underwent an initial cleaning process

187 records were received with 43 columns of the most diverse types of content and formats.

Four of them required interpretation work or using the IDs to search for the characteristics

- From the '**rocket**' we would like to learn the **booster name**
- From the '**payload**' we would like to learn the **mass of the payload** and the **orbit** that it is going to
- From the '**launchpad**' we would like to know the **name of the launch site** being used, the **longitude**, and the **latitude**.
- From the '**cores**' we would like to learn the outcome of the landing, the type of the landing, number of flights with that core, whether gridfins were used, whether the core is reused, whether legs were used, the landing pad used, the block of the core which is a number used to separate version of cores, the number of times this specific core has been reused, and the serial of the core.

id		rocket	payloads		launchpad	cores	flight_number	date_utc	date
0	5e9d0d95eda69955f709d1eb	5eb0e4b5b6c3bb0006eeb1e1	5e9e4502f5090995de566f86		{'core': '5e9e289df35918033d3b2623', 'flight': 1, 'gridfins': False, 'legs': False, 'reused': False, 'landing_attempt': False, 'landing_success': None, 'landing_type': None, 'landpad': None}		1	2006-03-24T22:30:00.000Z	2006-03-24
1	5e9d0d95eda69955f709d1eb	5eb0e4b6b6c3bb0006eeb1e2	5e9e4502f5090995de566f86		{'core': '5e9e289ef35918416a3b2624', 'flight': 1, 'gridfins': False, 'legs': False, 'reused': False, 'landing_attempt': False, 'landing_success': None, 'landing_type': None, 'landpad': None}		2	2007-03-21T01:10:00.000Z	2007-03-21
3	5e9d0d95eda69955f709d1eb	5eb0e4b7b6c3bb0006eeb1e5	5e9e4502f5090995de566f86		{'core': '5e9e289ef3591855dc3b2626', 'flight': 1, 'gridfins': False, 'legs': False, 'reused': False, 'landing_attempt': False, 'landing_success': None, 'landing_type': None, 'landpad': None}		4	2008-09-28T23:15:00.000Z	2008-09-28
4	5e9d0d95eda69955f709d1eb	5eb0e4b7b6c3bb0006eeb1e6	5e9e4502f5090995de566f86		{'core': '5e9e289ef359184f103b2627', 'flight': 1, 'gridfins': False, 'legs': False, 'reused': False, 'landing_attempt': False, 'landing_success': None, 'landing_type': None, 'landpad': None}		5	2009-07-13T03:35:00.000Z	2009-07-13
5	5e9d0d95eda69973a809d1ec	5eb0e4b7b6c3bb0006eeb1e7	5e9e4501f509094ba4566f84		{'core': '5e9e289ef359185f2b3b2628', 'flight': 1, 'gridfins': False, 'legs': False, 'reused': False, 'landing_attempt': False, 'landing_success': None, 'landing_type': None, 'landpad': None}		6	2010-06-04T18:45:00.000Z	2010-06-04

Selected columns, along with these, formed the initial attribute base for predicting success or failure

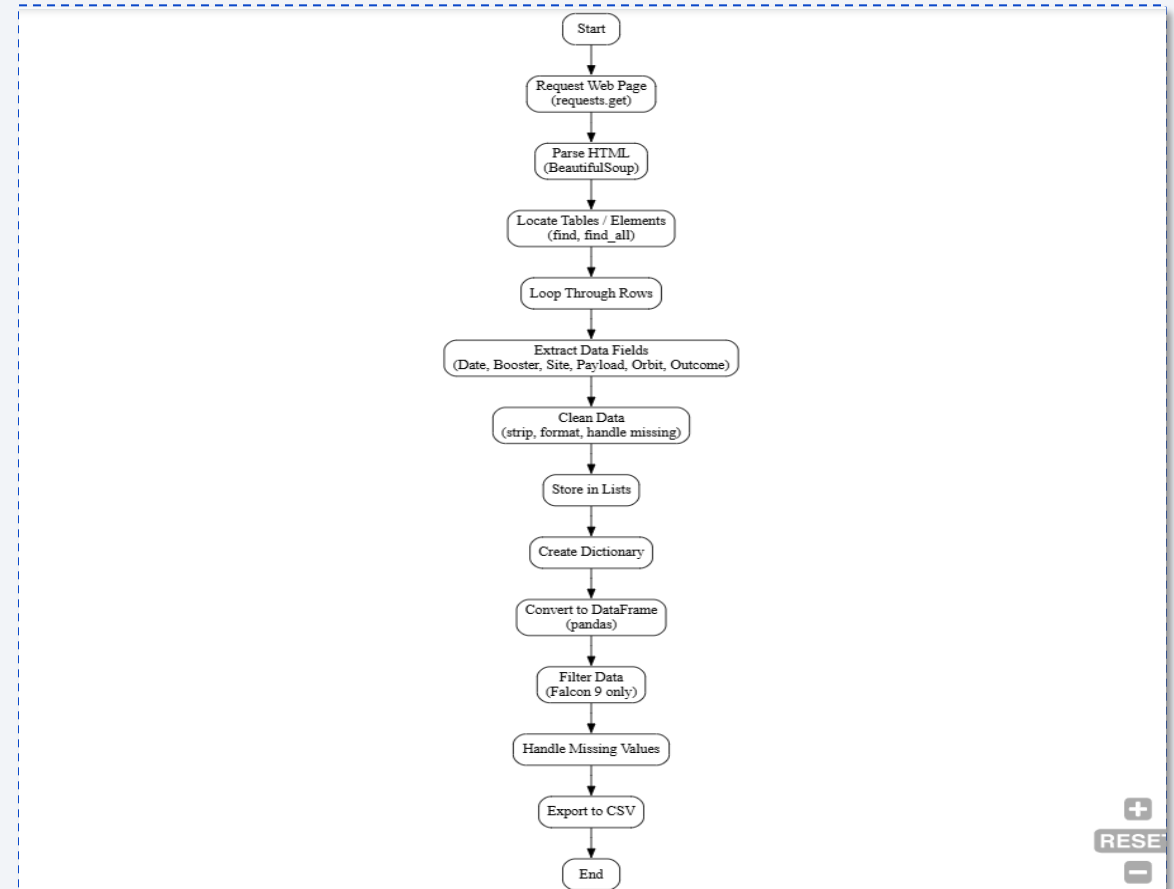
<u>Field</u>	Data Type	Description	Example
FlightNumber	Integer	Sequential number of the launch	1, 2, 3
Date	Date (string/date)	Launch date	2020-05-30
BoosterVersion	String	Version or identifier of the rocket booster	Falcon 9 Block 5
PayloadMass	Float	Payload mass in kilograms	5496.0
Orbit	String	Target orbit type of the mission	LEO, GTO, ISS
LaunchSite	String	Launch site location	KSC LC-39A
Outcome	String	First stage landing outcome	True ASDS, False RTLS
Flights	Integer	Number of previous flights of the booster	1, 2, 3
GridFins	Boolean	Indicates whether grid fins were used (aerodynamic control surfaces)	True / False
Reused	Boolean	Indicates whether the booster was reused	True / False
Legs	Boolean	Indicates whether the booster has landing legs	True / False
LandingPad	String	Landing pad used for recovery	LZ-1, ASDS OCISLY
Block	Float / Integer	Falcon 9 block version	5.0
ReusedCount	Integer	Number of times the booster has been reused	0, 1, 2
Serial	String	Unique booster identifier	B1049
Longitude	Float	Longitude of the launch site	-80.5772
Latitude	Float	Latitude of the launch site	28.5619

Additionally, an inspection showed that some rows in the dataset contain missing values.

- For the missing PayloadMass values, the mean value of the PayloadMass column will be calculated.*
- The LandingPad column will retain the value None to represent cases where landing pads were not used.*

Data Wrangling – Wiki page

- Web scraping was performed to collect historical records of Falcon 9 launches from a Wikipedia page titled "List of Falcon 9 and Falcon Heavy launches"
- Request the Falcon9 Launch Wiki page from its URL with Beatifull Soup
- Extract all column/variable names from the HTML table header
- Create a data frame by parsing the launch HTML tables



by <https://dreampuf.github.io/>

Web scraping was also performed to collect historical records of Falcon 9 launches from a Wikipedia page titled "List of Falcon 9 and Falcon Heavy launches"

2020 [\[edit \]](#)

In late 2019, [Gwynne Shotwell](#) stated that SpaceX hoped for as many as 24 launches for Starlink satellites in 2020,^[490] in addition to 14 or 15 non-Starlink launches. At 26 launches, 13 of which for Starlink satellites, Falcon 9 had its most prolific year, and Falcon rockets were second most prolific rocket family of 2020, only behind China's [Long March](#) rocket family.^[491]

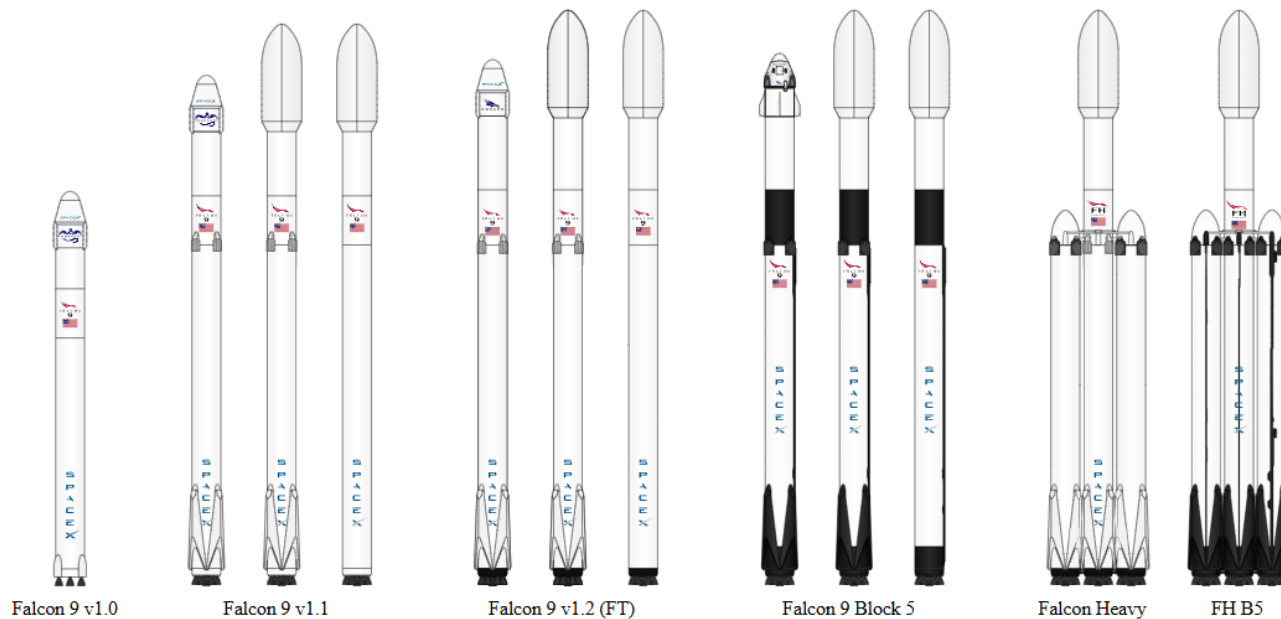
[hide] Flight No.	Date and time (UTC)	Version, Booster ^[b]	Launch site	Payload ^[c]	Payload mass	Orbit	Customer	Launch outcome	Booster landing
78	7 January 2020, 02:19:21 ^[492]	F9 B5 △ B1049.4	CCAFS, SLC-40	Starlink 2 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Success (drone ship)
Third large batch and second operational flight of Starlink constellation. One of the 60 satellites included a test coating to make the satellite less reflective, and thus less likely to interfere with ground-based astronomical observations. ^[493]									
79	19 January 2020, 15:30 ^[494]	F9 B5 △ B1046.4	KSC, LC-39A	Crew Dragon in-flight abort test ^[495] (Dragon C205.1)	12,050 kg (26,570 lb)	Sub-orbital ^[496]	NASA (CTS) ^[497]	Success	No attempt
An atmospheric test of the Dragon 2 abort system after Max Q . The capsule fired its SuperDraco engines, reached an apogee of 40 km (25 mi), deployed parachutes after reentry, and splashed down in the ocean 31 km (19 mi) downrange from the launch site. The test was previously slated to be accomplished with the Crew Dragon Demo-1 capsule; ^[498] but that test article exploded during a ground test of SuperDraco engines on 20 April 2019. ^[419] The abort test used the capsule originally intended for the first crewed flight. ^[499] As expected, the booster was destroyed by aerodynamic forces after the capsule aborted. ^[500] First flight of a Falcon 9 with only one functional stage — the second stage had a mass simulator in place of its engine.									
80	29 January 2020, 14:07 ^[501]	F9 B5 △ B1051.3	CCAFS, SLC-40	Starlink 3 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Success (drone ship)
Third operational and fourth large batch of Starlink satellites, deployed in a circular 290 km (180 mi) orbit. One of the fairing halves was caught, while the other was fished out of the ocean. ^[502]									
81	17 February 2020, 15:05 ^[503]	F9 B5 △ B1056.4	CCAFS, SLC-40	Starlink 4 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Failure (drone ship)
Fourth operational and fifth large batch of Starlink satellites. Used a new flight profile which deployed into a 212 km × 386 km (132 mi × 240 mi) elliptical orbit instead of launching into a circular orbit and firing the second stage engine twice. The first stage									



	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version	Booster	Booster landing	Date	Time
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	F9 v1.07B0003.18		Failure	4 June 2010	18:45
1	2	CCAFS	Dragon	0	LEO		Success	F9 v1.07B0004.18		Failure	8 December 2010	15:43
2	3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.07B0005.18		No attempt	22 May 2012	07:44
3	4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success	F9 v1.07B0006.18		No attempt	8 October 2012	00:35
4	5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success	F9 v1.07B0007.18		No attempt	1 March 2013	15:10

1. Request the Falcon9 Launch Wiki page from its URL with Beatifull Soup
2. Extract all column/variable names from the HTML table header
3. Create a data frame by parsing the launch HTML tables

To better focus the analysis, only Falcon 9 launches were selected



Removing Falcon 1 launches from analyses focused on SpaceX's current success (keeping only the Falcon 9) is a practice to ensure the accuracy of operational and financial data for several fundamental reasons:

Difference in Purpose and Scale: The Falcon 1 was a development vehicle ("minimum viable product") to test components, such as the Merlin engine. The Falcon 9, on the other hand, is a heavy-lift, operational rocket. Including the Falcon 1 distorts the company's reliability and payload capacity metrics.

Testing Phase vs. Operational Phase: The first Falcon 1 launches failed (3 failures in a row), almost leading the company to bankruptcy. They served to learn, not to consistently transport commercial payloads, unlike the Falcon 9, which has become one of the most reliable and widely used rockets in history.

Focus on Reusability: The Falcon 1 was not designed for reuse, while the Falcon 9 Block 5 (the current standard) is focused on rapid and economic reuse, drastically changing the cost per launch.

Profit Margin and Market: SpaceX concluded that the market for small payloads (Falcon 1) was less profitable, abandoning it to focus resources on the Falcon 9, which offers higher financial returns.

In summary, the Falcon 1 is considered a historic test vehicle, while the Falcon 9 is the current commercial vehicle. Removing the Falcon 1 makes the analysis of SpaceX's current efficiency more realistic.

From Wikipedia and IA

Important relationships were evaluated using the Pandas and Matplotlib libraries

EDA with Data Visualization:

- FlightNumber vs. PayloadMass and overlay the outcome of the launch
- Relationship between Flight Number and Launch Site
- Relationship between Payload and Launch Site
- Relationship between Success Rate of each Orbit type
- Relationship between FlightNumber and Orbit type
- Relationship between Payload and Orbit type
- Visualize the launch success yearly trend

Relevant business measurements were obtained using queries via the sqlalchemy, ipython-sql, and prettytable libraries

EDA with SQL

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA'
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display the launch count per customer, showing the most frequent ones
- Display the payload mass carried by the frequent customers versus carried by the others
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first succesful landing outcome in ground pad was achieved
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List all the booster_versions that have carried the maximum payload mass
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

[https://github.com/CarlosJunior41/Space_Y_project/blob/main/jupyter-labs-eda-sql-coursera_sqlite%20\(1\).ipynb](https://github.com/CarlosJunior41/Space_Y_project/blob/main/jupyter-labs-eda-sql-coursera_sqlite%20(1).ipynb)

The success rate of a launch depends on many factors, including the initial position of the rocket's trajectory

Points of interest were marked on the map using the folium, wget, and pandas libraries

- Add each site's location on a map using site's latitude and longitude coordinates
- Creating markers indicating launch results for each site to see which sites have high success rates
- For each site, distances to the sea and other points of interest, such as roads and dwellings, were studied

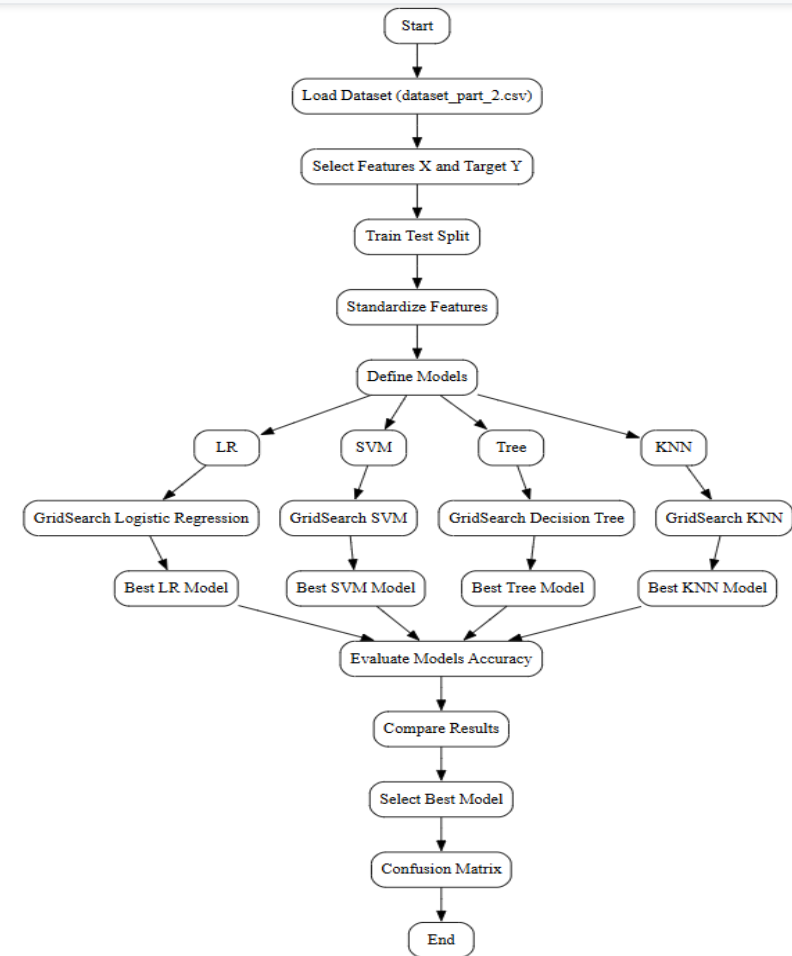
To obtain more information from the SpaceX dataset, a dashboard was created that displays it more easily than static graphs

Creating a dashboard using the pandas, dash, and plotly.express libraries with:

- Creating a drop-down input component to select a specific launch site to check its detailed success rate
- Creating a callback function to render a success pie chart based on the selected website's dropdown menu
- Additionally, creating a bar graph to show the number of successful launches of the different F9 Booster versions
- An interval slider has been added to select the payload launched by the site indicated in the dropdown selector
- Creating a scatter plot with the x-axis representing the payload and the y-axis representing the launch result (i.e., the class column). In this way, it is possible to visually observe how the payload may be correlated with the mission results for the selected site launch

Predictive Analysis (Classification)

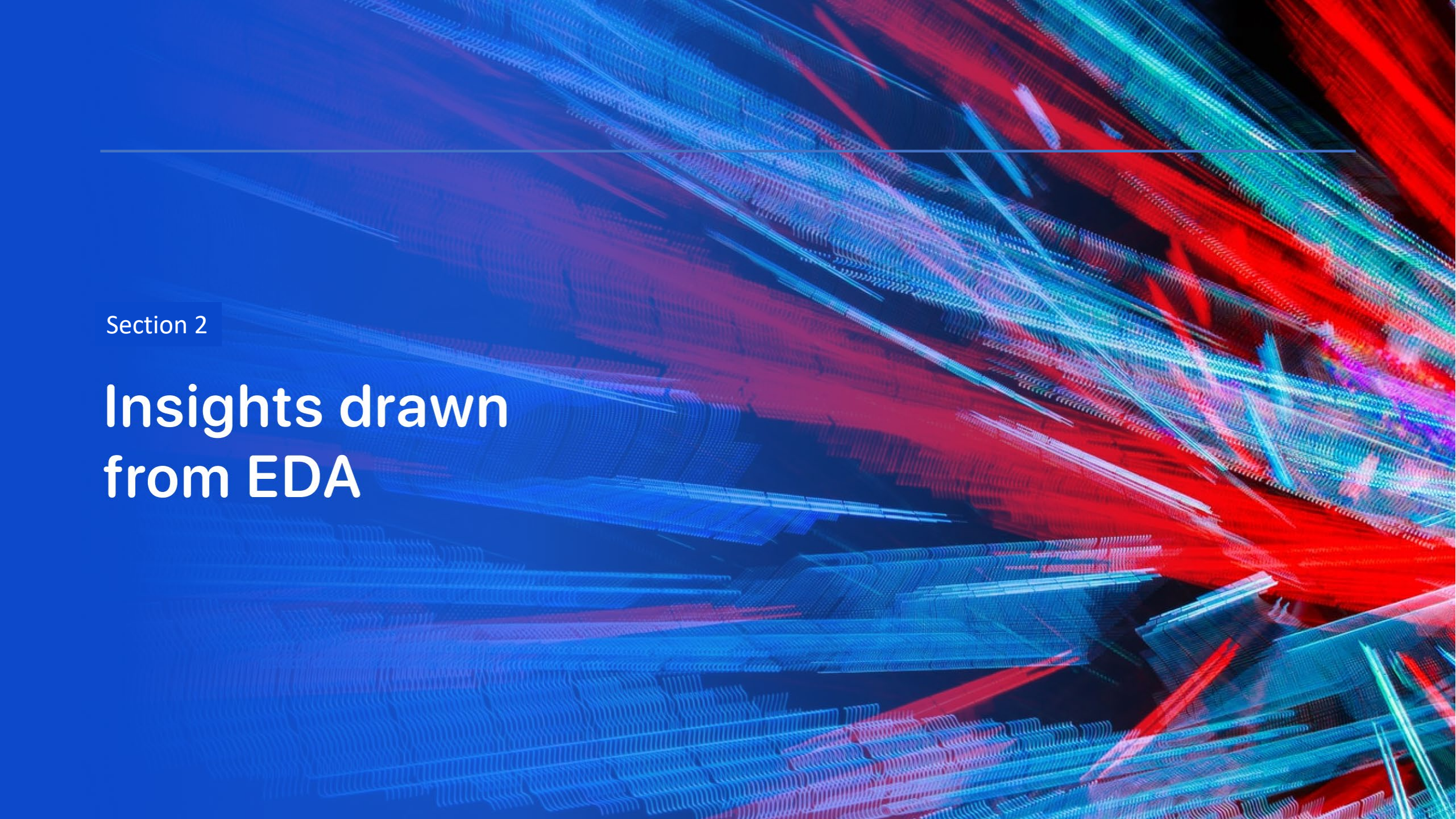
- The resulting database underwent a process of isolating the target variable from the predictor variables, followed by normalization or standardization (-score) of the features
- The next step is to divide the data into training and testing databases
- To optimize performance, the data underwent a hyperparameter tuning process for four distinct models: SVM, KNN, Classification Trees, and Logistic Regression
- Analyzing the confusion matrix and metrics of these models' results indicates the most suitable one and may suggest improvements



by <https://dreampuf.github.io/>

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results
- Proposed formulas for determining costs

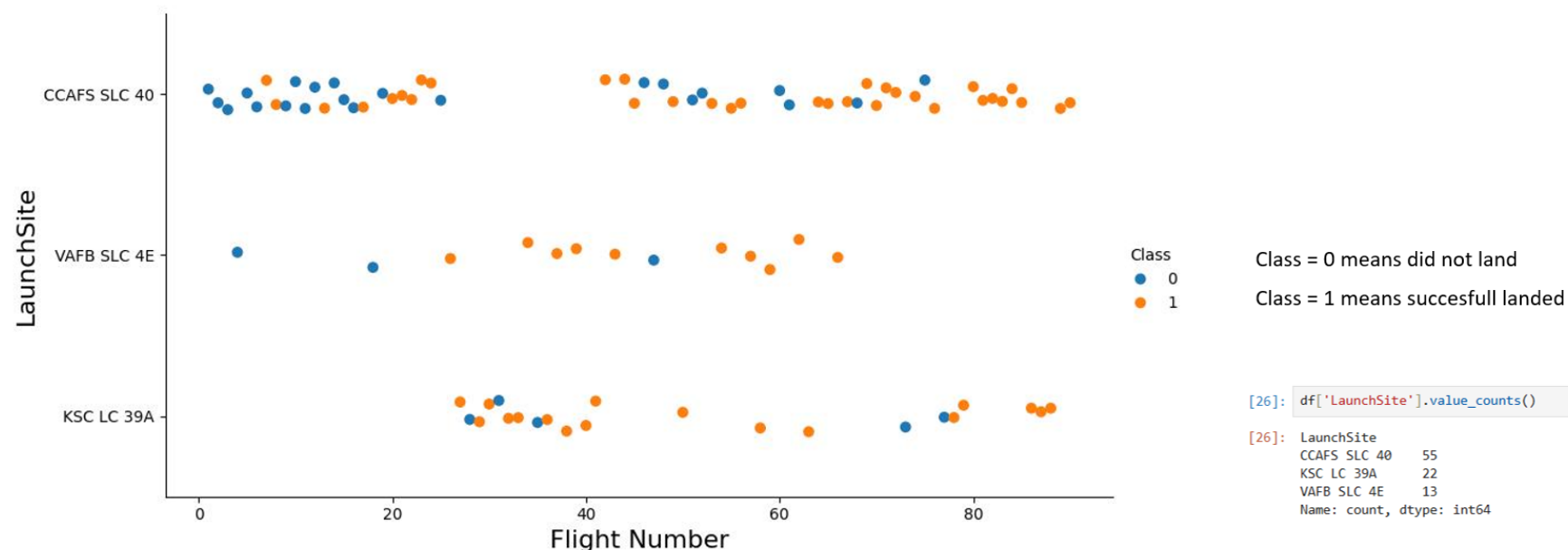
The background of the slide is a dark blue gradient. It is overlaid with numerous diagonal streaks and bands of light blue and red, creating a sense of motion or data flow. A thin, solid white horizontal line spans the width of the slide, positioned approximately one-third of the way down from the top.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Ninety launches were conducted between June 2010 and May 2020, with a first-stage recovery success rate of 67%.

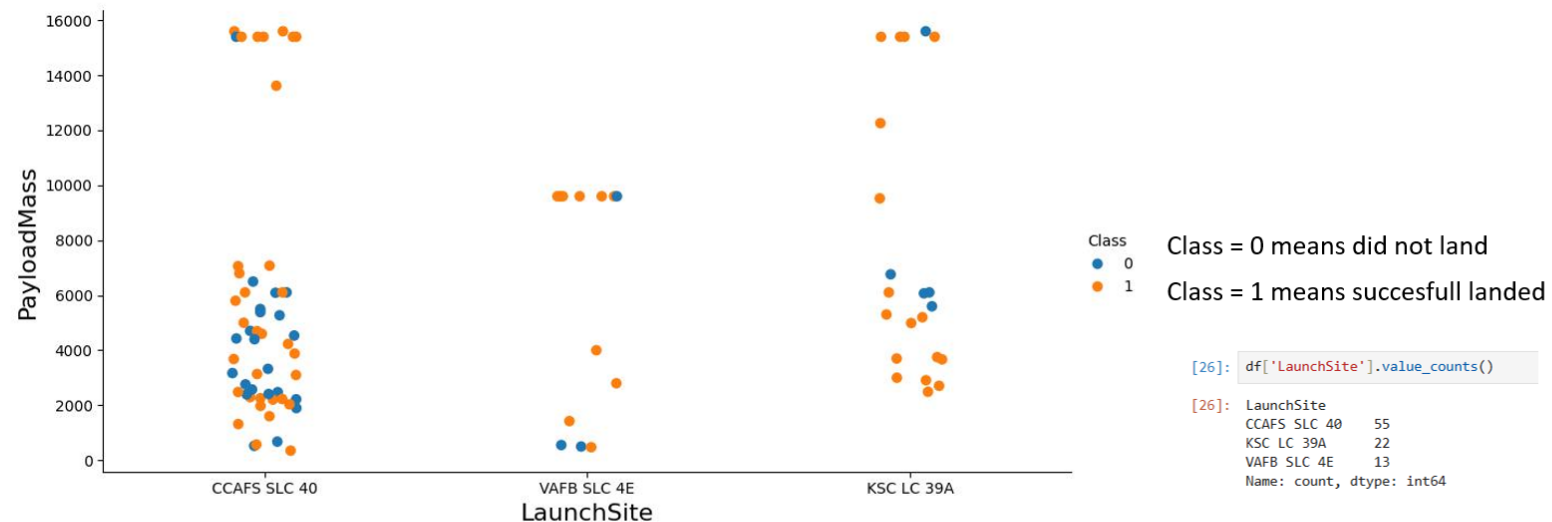


The launch success rate may depend on many factors such as payload mass, orbit type, and so on. It may also depend on the location and proximities of a launch site, i.e., the initial position of rocket trajectories.

Finding an optimal location for building a launch site certainly involves many factors and hopefully we could discover some of the factors by analyzing the existing launch site locations.

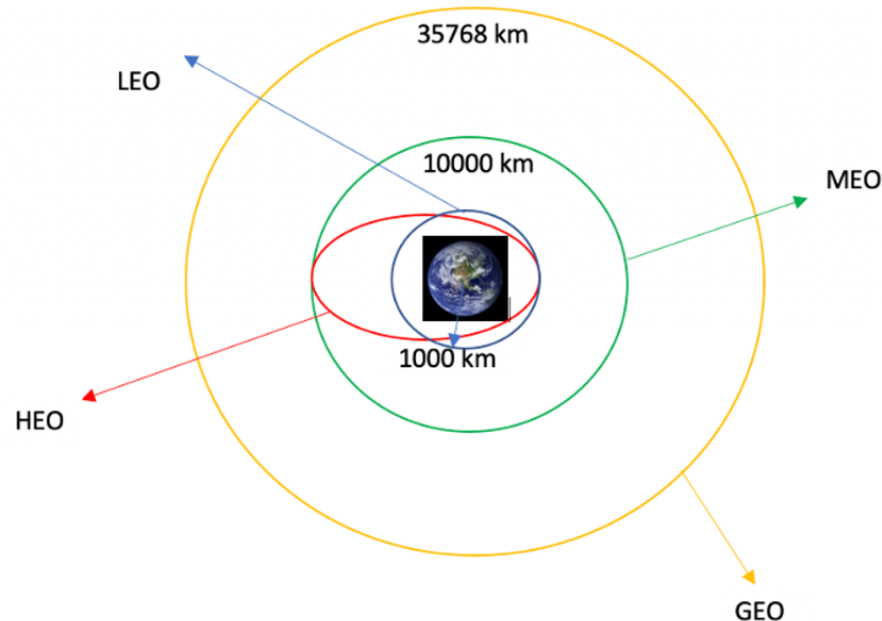
Payload vs. Launch Site

The VAFB-SLC launchsite there are no rockets launched for heavypayload mass (greater than 10000)



Orbit Types

There are various orbits with different purposes that can be reached by launches



Each launch aims to an dedicated orbit, and here are some common orbit types

LEO: Low Earth orbit (LEO) is an Earth-centred orbit with an altitude of 2,000 km (1,200 mi) or less (approximately one-third of the radius of Earth), [1] or with at least 11.25 periods per day (an orbital period of 128 minutes or less) and an eccentricity less than 0.25. [2] Most of the manmade objects in outer space are in LEO [1].

VLEO: Very Low Earth Orbits (VLEO) can be defined as the orbits with a mean altitude below 450 km. Operating in these orbits can provide a number of benefits to Earth observation spacecraft as the spacecraft operates closer to the observation [2].

GTO (Geostationary Transfer Orbit): A geostationary transfer orbit is an elliptical Earth orbit used to transfer satellites from Low Earth orbit (LEO) to geostationary orbit (GEO). In a GTO, the perigee (closest point to Earth) is much lower than GEO altitude, while the apogee (farthest point) reaches approximately 22,236 miles (35,786 kilometers) above Earth's equator — the altitude of a geostationary orbit. Satellites in GTO use onboard propulsion to circularize their orbit at GEO altitude, where they can provide services such as weather monitoring, communications, and surveillance. [3].

SSO (or SO): It is a Sun-synchronous orbit also called a heliosynchronous orbit is a nearly polar orbit around a planet, in which the satellite passes over any given point of the planet's surface at the same local mean solar time [4].

ES-L1: At the Lagrange points the gravitational forces of the two large bodies cancel out in such a way that a small object placed in orbit there is in equilibrium relative to the center of mass of the large bodies. L1 is one such point between the sun and the earth [5].

HEO A highly elliptical orbit, is an elliptical orbit with high eccentricity, usually referring to one around Earth [6].

ISS A modular space station (habitable artificial satellite) in low Earth orbit. It is a multinational collaborative project between five participating space agencies: NASA (United States), Roscosmos (Russia), JAXA (Japan), ESA (Europe), and CSA (Canada) [7].

MEO Geocentric orbits ranging in altitude from 2,000 km (1,200 mi) to just below geosynchronous orbit at 35,786 kilometers (22,236 mi). Also known as an intermediate circular orbit. These are "most commonly at 20,200 kilometers (12,600 mi), or 20,650 kilometers (12,830 mi), with an orbital period of 12 hours [8].

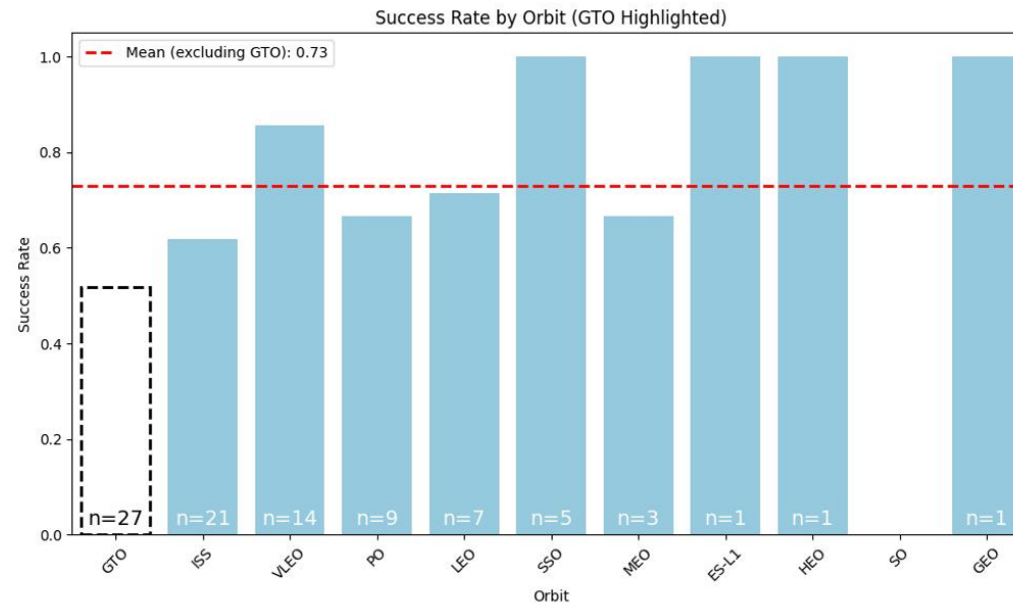
HEO Geocentric orbits above the altitude of geosynchronous orbit (35,786 km or 22,236 mi) [9].

GEO It is a circular geosynchronous orbit 35,786 kilometres (22,236 miles) above Earth's equator and following the direction of Earth's rotation [10].

PO It is one type of satellites in which a satellite passes above or nearly above both poles of the body being orbited (usually a planet such as the Earth [11]).

Success Rate vs. Orbit Type

The weighted average of the different success rates associated with these orbits results in 73%

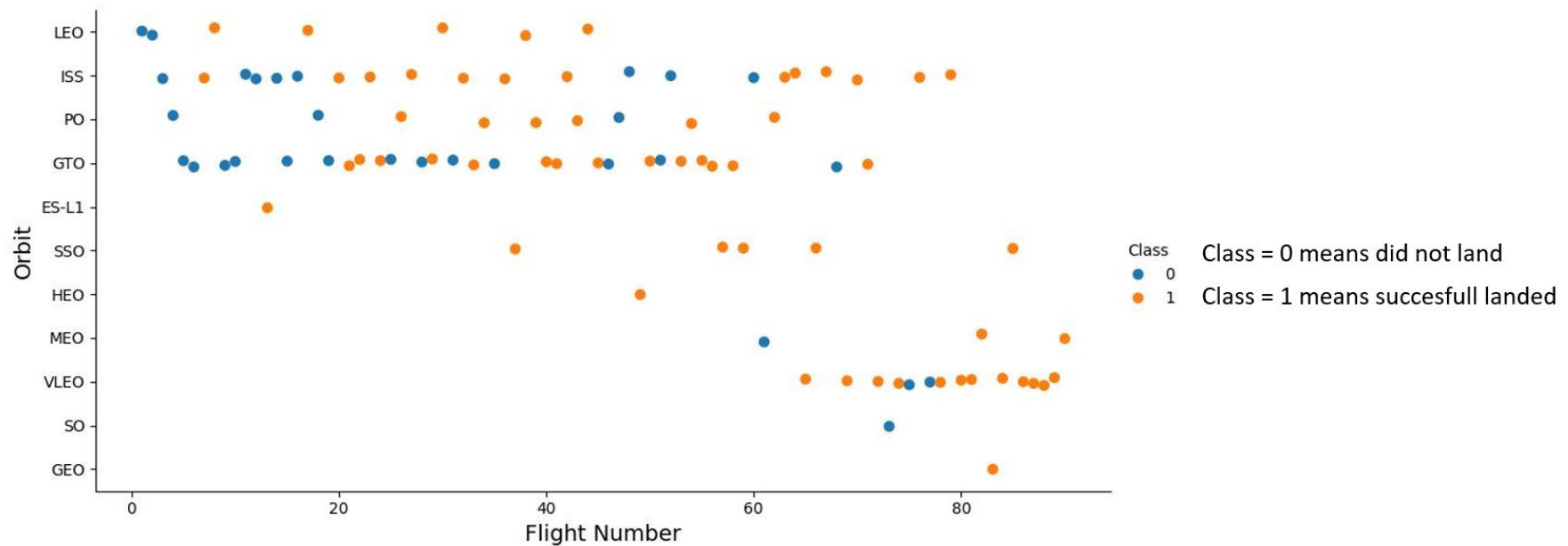


Twenty-seven launches to geostationary transfer orbit (GTO) were not considered because it is a transfer orbit and not a geostationary orbit in itself.

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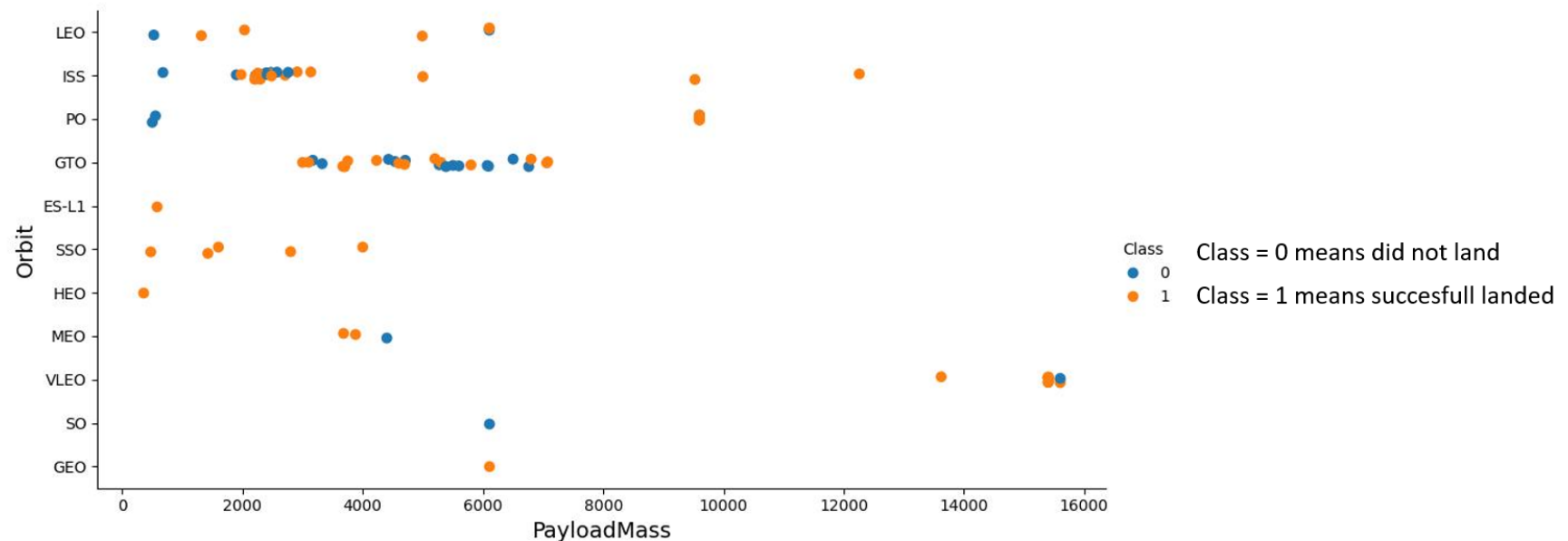
Flight Number vs. Orbit Type

As the company progressed, other orbits were introduced as launch targets



Payload vs. Orbit Type

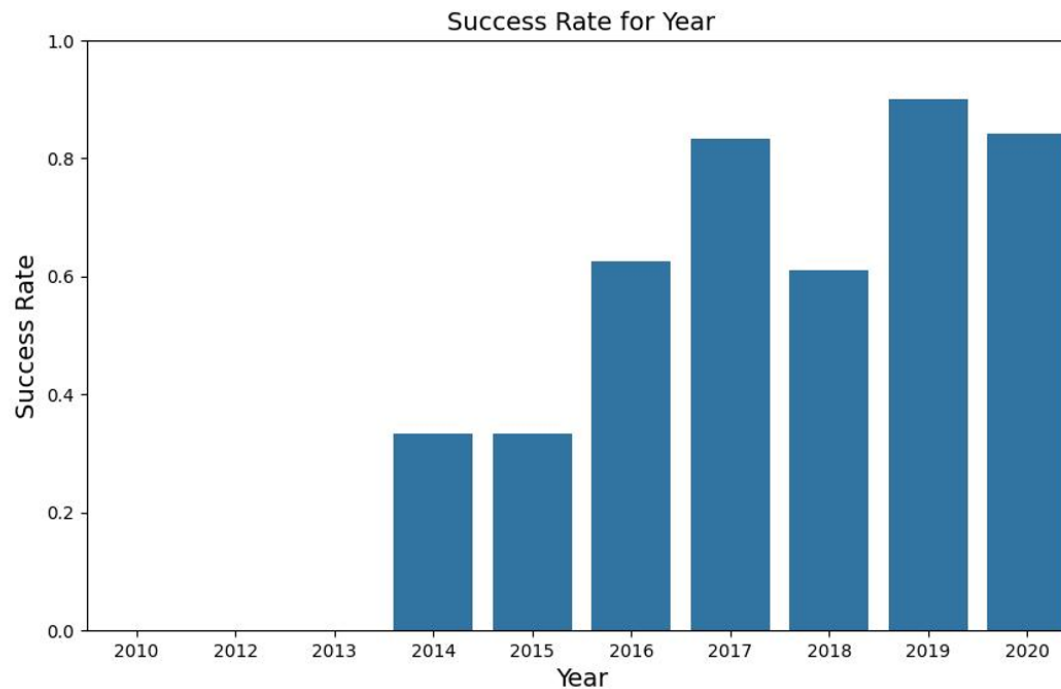
The successful or positive landing rate is higher for Polar, LEO, and ISS orbits.



For geostationary transfer orbits (GTOs), sought several times during the analysis period, we cannot clearly distinguish between them, as both successful and unsuccessful landings (unsuccessful missions) occur in this region.

Launch Success Yearly Trend

The annual trend of successful launches has been increasing

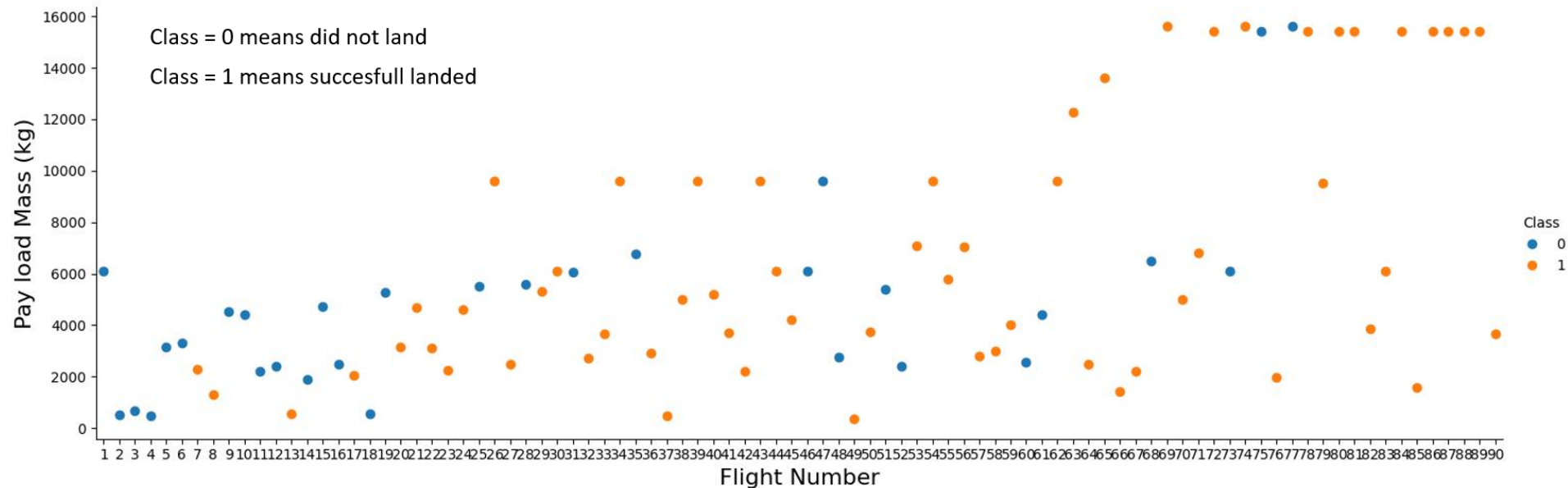


- The success rate since 2013 kept increasing till 2017 (stable in 2014) and after 2015 it started increasing

Payload vs. Flight Number

The sequence of flight numbers related to payload mass shows how the success rate in first-stage recovery has progressed

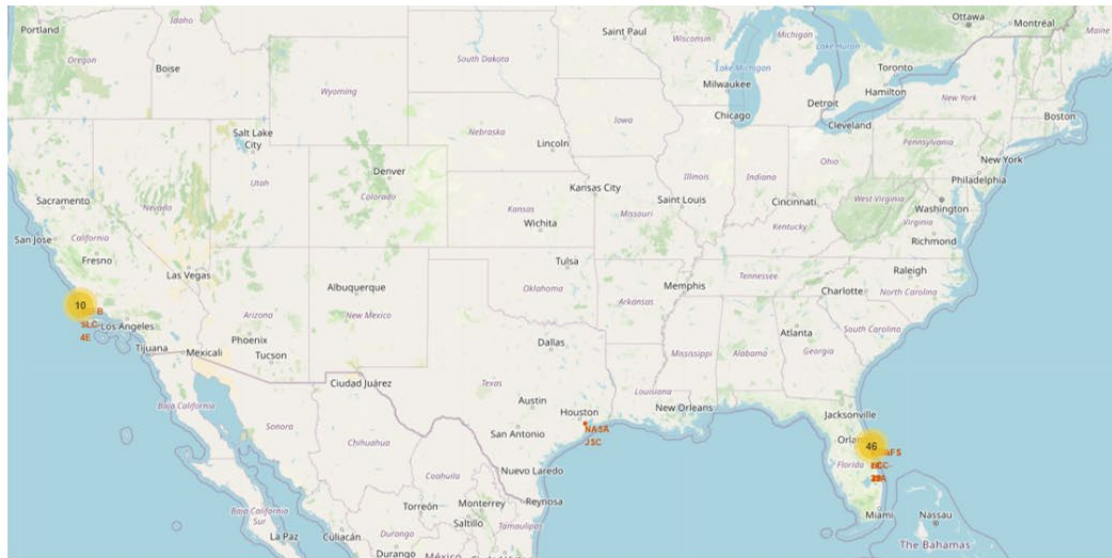
With the increasing success rate, the weight transported also grew



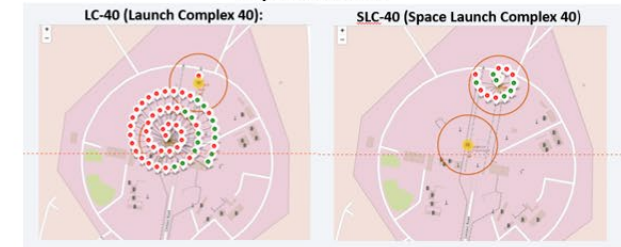
All Launch Site Names

These launches were carried out from three launch sites, two in Florida and one in California

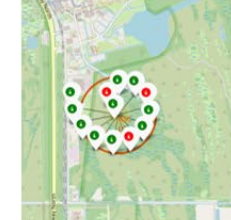
Vandenberg Space Force Base



Cape Canaveral



Kennedy Space Center



Launches began at the **Cape Canaveral LC-40 (Launch Complex 40)** launch site, which carried out almost all of SpaceX's first 26 launches

In the period between the 26th and 40th launches, the CCAFS launch site did not carry out any launches. Launches were carried out at the neighboring **Kennedy Space Center (KSC)** launch site, with 11 launches, 8 of which were successful, and at the **Vandenberg Space Force Base (VALFB)** launch site, which carried out 5 successful launches.

From the 40th launch onwards the **Cape Canaveral SLC-40 (Space Launch Complex 40)** Launch Site resumed operations with 33 launches.

Launch Site Names Begin with 'CCA'

“CCA” is short for Cape Canaveral, the location of the two main launch sites: LC40 and SLC40

Display 5 records where launch sites begin with the string 'CCA'

```
[12]: %%sql
      SELECT * FROM SPACEXTABLE WHERE Launch_site LIKE 'CCA%' LIMIT 5;

* sqlite:///my_data1.db
Done.
```

```
[12]:
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

CCA FS LC-40 (Launch Complex 40): This was the original name of the complex used by the US Air Force during the 1960s to 2000s for Titan III and Titan IV rockets.

CCA FS SLC-40 (Space Launch Complex 40): This is the current name of the facility after being leased by SpaceX in 2007, rebuilt and upgraded to launch the Falcon 9 rocket.

Frequent Customers and others

51 flights for NASA and SpaceX (either directly or in partnership)

Customer	qtd
NASA (CRS)	20
SpaceX	13
SpaceX, Planet Labs	2
U.S. Air Force NASA NOAA	1
SpaceX, Spaceflight Industries (BlackSky), Planet Labs	1
SpaceX, Planet Labs, PlanetIQ	1
NASA / NOAA / ESA / EUMETSAT	1
NASA (LSP) NOAA CNES	1
NASA (LSP)	1
NASA (CTS)	1
NASA (CRS), Kacific 1	1
NASA (COTS) NRO	1
NASA (COTS)	1
NASA (CCP)	1
NASA (CCDev)	1
NASA (CCD)	1
Iridium Communications GFZ , NASA	1
Hisdesat exactEarth SpaceX	1
CONAE, PlanetIQ, SpaceX	1

50 flights distributed among 34 other clients

Customer	qtd
Iridium Communications	7
SES	5
Thaicom	2
Telesat	2
SKY Perfect JSAT Group	2
Orbcomm	2
AsiaSat	2
ABS Eutelsat	2
USSF	1
USAF	1
U.S. Space Force	1
U.S. Air Force	1
Turkmenistan National Space Agency	1
Thales-Alenia/BTRC	1
Telkom Indonesia	1
Spaceflight Industries	1
Spacecom	1
Sky Perfect JSAT, Kacific 1	1
SES EchoStar	1
Republic of Korea Army, Spaceflight Industries (BlackSky)	1
PSN, Spacell / IAI	1
Northrop Grumman	1

Customer	qtd
NSPO	1
NRO	1
MDA	1
KT Corporation	1
Intelsat	1
Inmarsat	1
Hispasat NovaWurks	1
Es hailSat	1
EchoStar	1
Canadian Space Agency (CSA)	1
CONAE	1
Bulsatcom	1

Payload from frequent customers and the others

**51 flights for NASA and SpaceX
(itself or in association)**

```
%%sql
SELECT SUM(PAYLOAD_MASS_KG_)
FROM SPACEXTABLE
WHERE Customer LIKE '%NASA%' OR Customer LIKE '%SpaceX%';
```

```
* sqlite:///my_data1.db
Done.
```

SUM(PAYLOAD_MASS_KG_)
358892

50 flights distributed among 34 clients

```
%%sql
SELECT SUM(PAYLOAD_MASS_KG_)
FROM SPACEXTABLE
WHERE Customer NOT LIKE '%NASA%' AND Customer NOT LIKE '%SpaceX%';
```

```
* sqlite:///my_data1.db
Done.
```

SUM(PAYLOAD_MASS_KG_)
261075

$$358.892 + 261.075 = 619.967$$

Frequent customers account for 50% of launches and carry 58% of the total load

Total Payload Mass

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[13]: %%sql
      SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';

      * sqlite:///my_data1.db
      Done.

[13]: SUM(PAYLOAD_MASS__KG_)
      45596
```

NASA sent 46 tons into space out of a total payload of 620 tons, i.e., 7%.

Average Payload Mass by F9 v1.1 and F9 B5 booster version

Display average and max payload mass carried by booster version F9 v1.1

```
] : %%sql
SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%';
* sqlite:///my_data1.db
Done.
:] : AVG(PAYLOAD_MASS_KG_)
      2534.6666666666665

:] : %%sql
SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%';
* sqlite:///my_data1.db
Done.
:] : MAX(PAYLOAD_MASS_KG_)
      4707
```

With the F9 v1.1 booster, the average loaded weight represented 58% of the maximum weight.

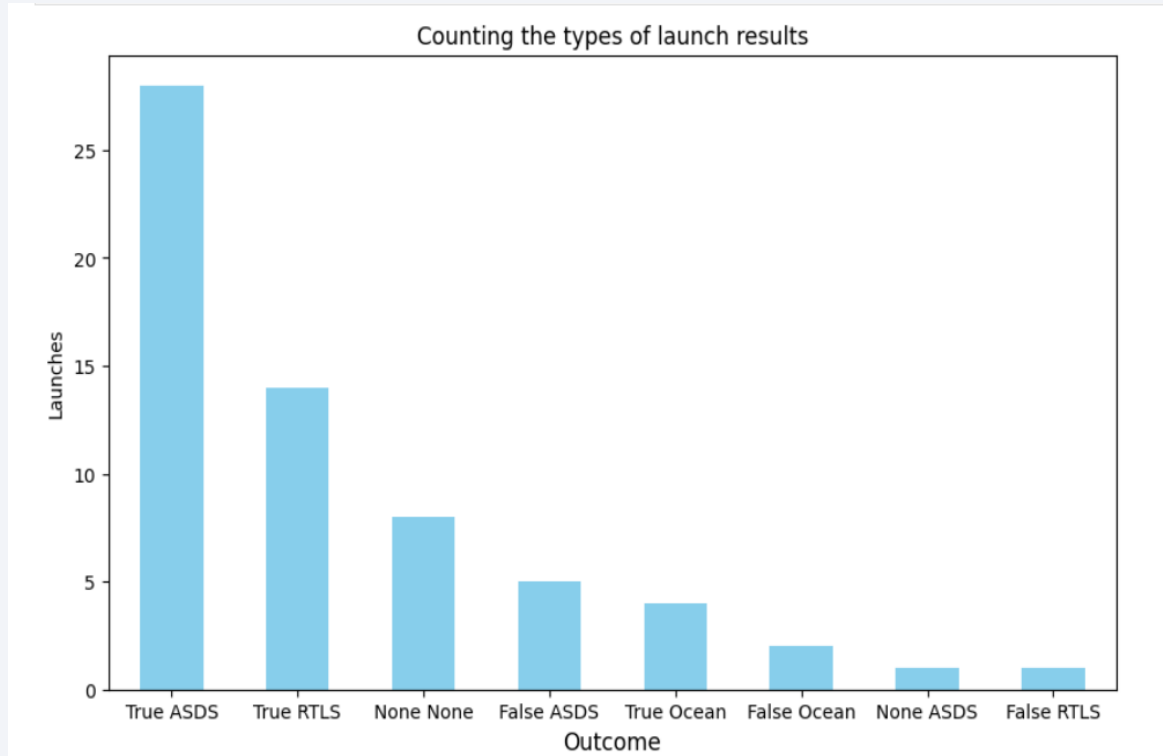
Display average and max payload mass carried by booster version F9 B5 B

```
: %%sql
SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 B5 B%';
* sqlite:///my_data1.db
Done.
: AVG(PAYLOAD_MASS_KG_)
      10474.354838709678

: %%sql
SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 B5 B%';
* sqlite:///my_data1.db
Done.
: MAX(PAYLOAD_MASS_KG_)
      15600
```

With the F9 B5 B booster, the average weight represented 67% of the heaviest weight carried

Successful recovery from the first stage can occur on a drone ship, on the ground pad, or in a specific region of the oceans



True Ocean means the mission outcome was successfully landed to a specific region of the ocean while

False Ocean means the mission outcome was unsuccessfully landed to a specific region of the ocean.

True RTLS means the mission outcome was successfully landed to a ground pad

False RTLS means the mission outcome was unsuccessfully landed to a ground pad.

True ASDS means the mission outcome was successfully landed to a drone ship

False ASDS means the mission outcome was unsuccessfully landed to a drone ship.

None ASDS and **None None** these represent a failure to land.

The success indicator label is created from the selection of successful outcome types

First Successful Ground Landing Date

KPI's Indicators

List the total number of successful and failure mission outcomes

```
%%sql ***
```

```
* sqlite:///my_data1.db
```

Done.

Mission_Outcome	Count
Success	98
Success (payload status unclear)	1
Success	1
Failure (in flight)	1

List the total number of successful and failure Landings Outcomes

```
%%sql ***
```

```
* sqlite:///my_data1.db
```

Done.

Landing_Outcome	Count
Success	38
No attempt	21
Success (drone ship)	14
Success (ground pad)	9
Failure (drone ship)	5
Controlled (ocean)	5
Failure	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1
No attempt	1

Mission Outcome: Refers to the main objective of the launch:

- Placing the satellite/payload into the correct orbit
- Successfully testing a system
- Completing the planned mission

Main indicator for the client paying for the launch

Landing Outcome:

Refers to the return of the first stage of the rocket (booster):

- Whether it successfully landed (on land or droneship)
- Whether it failed (exploded, fell in to the sea, etc.)

Internal company indicator (cost reduction through reuse)

Milestone dates

List the date when the first successful landing outcome was achieved.

```
%%sql
```

```
SELECT MIN(Date) FROM SPACEXTABLE WHERE Mission_Outcome = 'Success';
```

```
* sqlite:///my_data1.db
```

Done.

MIN(Date)

2010-06-04

```
%%sql
```

```
SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

Done.

MIN(Date)

2015-12-22

```
%%sql
```

```
SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)';
```

```
* sqlite:///my_data1.db
```

Done.

MIN(Date)

2016-04-08

```
%%sql
```

```
SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success';
```

```
* sqlite:///my_data1.db
```

Done.

MIN(Date)

2018-07-22

Successful Drone Ship Landing with Payload between 4000 and 6000

List the name, payload and date of the boosters which have success in drone ship

```
%%sql ...
```

```
* sqlite:///my_data1.db  
Done.
```

Booster_Version	PAYLOAD_MASS_KG_	Date
F9 FT B1021.1	3136	2016-04-08
F9 FT B1022	4696	2016-05-06
F9 FT B1023.1	3100	2016-05-27
F9 FT B1026	4600	2016-08-14
F9 FT B1029.1	9600	2017-01-14
F9 FT B1021.2	5300	2017-03-30
F9 FT B1029.2	3669	2017-06-23
F9 FT B1036.1	9600	2017-06-25
F9 FT B1038.1	475	2017-08-24
F9 B4 B1041.1	9600	2017-10-09
F9 FT B1031.2	5200	2017-10-11
F9 B4 B1042.1	3500	2017-10-30
F9 B4 B1045.1	362	2018-04-18
F9 B5 B1046.1	3600	2018-05-11

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%%sql ...
```

```
* sqlite:///my_data1.db  
Done.
```

Booster_Version

F9 FT B1022	←
F9 FT B1026	←
F9 FT B1021.2	←
F9 FT B1031.2	←

The last digit after the decimal point in the SpaceX booster code (e.g., B1021.2) usually indicates the number of uses of that stage, with 1 being the first flight, 2 the second, and so on

It is noted that the reuse of the first stage is frequent for various load sizes

Successful Groun Pad Landing with Payload between 4000 and 6000

List the names of the boosters which have success in ground pad

```
%%sql ...
```

```
* sqlite:///my_data1.db  
Done.
```

Booster_Version	PAYLOAD_MASS_KG_	Date
F9 FT B1019	2034	2015-12-22
F9 FT B1025.1	2257	2016-07-18
F9 FT B1031.1	2490	2017-02-19
F9 FT B1032.1	5300	2017-05-01
F9 FT B1035.1	2708	2017-06-03
F9 B4 B1039.1	3310	2017-08-14
F9 B4 B1040.1	4990	2017-09-07
F9 FT B1035.2	2205	2017-12-15
F9 B4 B1043.1	5000	2018-01-08

List the names of the boosters which have success in ground pad and have payload mass greater than 4000 but less than 6000

```
%%sql ...
```

```
* sqlite:///my_data1.db  
Done.
```

```
Booster_Version  
  
F9 FT B1032.1  
F9 B4 B1040.1  
F9 B4 B1043.1
```

There is a noticeable decrease in the reuse of boosters aimed at landings on the ground pad

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

%%sql ...

```
* sqlite:///my_data1.db
Done.
```

Mission_Outcome	Count
Success	98
Success (payload status unclear)	1
Success	1
Failure (in flight)	1

The company achieved a 97% success rate in the 101 launch operations.

Excluding 21 launches where there was no intention of recovery, the company achieved a 76% recovery rate for boosters.

List the total number of successful and failure Landings Outcomes

%%sql ...

```
* sqlite:///my_data1.db
Done.
```

Landing_Outcome	Count
Success	38
No attempt	21
Success (drone ship)	14
Success (ground pad)	9
Failure (drone ship)	5
Controlled (ocean)	5
Failure	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1
No attempt	1

It is important to verify the degree of booster recovery achieved in these five landings classified as Controlled Ocean

Boosters Carried Maximum Payload

List all the booster_versions that have carried the maximum payload mass

```
%%sql ●●●
```

```
* sqlite:///my_data1.db
```

Done.

Booster_Version	Date	Mission_Outcome	Landing_Outcome
F9 B5 B1048:4	2019-11-11	Success	Success
F9 B5 B1049:4	2020-01-07	Success	Success
F9 B5 B1051:3	2020-01-29	Success	Success
F9 B5 B1056:4	2020-02-17	Success	Failure
F9 B5 B1048:5	2020-03-18	Success	Failure
F9 B5 B1051:4	2020-04-22	Success	Success
F9 B5 B1049:5	2020-06-04	Success	Success
F9 B5 B1060:2	2020-09-03	Success	Success
F9 B5 B1058:3	2020-10-06	Success	Success
F9 B5 B1051:6	2020-10-18	Success	Success
F9 B5 B1060:3	2020-10-24	Success	Success
F9 B5 B1049:7	2020-11-25	Success	Success

- It's impressive how many times these boosters are being reused
- This reusability should be an important indicator in cost calculations

2015 Launch Records

List the records which will display the month names, all failure landing_outcomes,booster versions,

```
%%sql ●●●
```

```
* sqlite:///my_data1.db  
Done.
```

month_name	Booster_Version	Launch_Site	PAYLOAD_MASS_KG_	Mission_Outcome	Landing_Outcome
January	F9 v1.1 B1012	CCAFS LC-40	2395	Success	Failure (drone ship)
February	F9 v1.1 B1013	CCAFS LC-40	570	Success	Controlled (ocean)
March	F9 v1.1 B1014	CCAFS LC-40	4159	Success	No attempt
April	F9 v1.1 B1015	CCAFS LC-40	1898	Success	Failure (drone ship)
April	F9 v1.1 B1016	CCAFS LC-40	4707	Success	No attempt
June	F9 v1.1 B1018	CCAFS LC-40	1952	Failure (in flight)	Precluded (drone ship)
December	F9 FT B1019	CCAFS LC-40	2034	Success	Success (ground pad)

List the records which will display the month names, failure landing_outcomes in drone ship

```
%%sql ●●●
```

```
* sqlite:///my_data1.db  
Done.
```

month_name	Booster_Version	Launch_Site	Mission_Outcome	Landing_Outcome
January	F9 v1.1 B1012	CCAFS LC-40	Success	Failure (drone ship)
April	F9 v1.1 B1015	CCAFS LC-40	Success	Failure (drone ship)

Despite a high mission success rate in 2015, the booster recovery rate was only 14%%

In the first half of 2015, the company focused on ocean landings. In addition to these two failures, there was also a controlled ocean landing and a flight that failed earlier but was intended for a drone ship.

This failed flight interrupted launches for the rest of the year until December when there was a successful recovery, but this time on a Ground Pad.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank the count of Mission Outcomes between the date 2010-06-04 and 2017-03-20

```
%%sql ●●●
```

```
* sqlite:///my_data1.db  
Done.
```

Mission_Outcome	Total_Count
-----------------	-------------

Success	30
---------	----

Failure (in flight)	1
---------------------	---

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20

```
%%sql ●●●
```

```
* sqlite:///my_data1.db  
Done.
```

Landing_Outcome	Total_Count
-----------------	-------------

No attempt	10
------------	----

Success (drone ship)	5
----------------------	---

Failure (drone ship)	5
----------------------	---

Success (ground pad)	3
----------------------	---

Controlled (ocean)	3
--------------------	---

Uncontrolled (ocean)	2
----------------------	---

Failure (parachute)	2
---------------------	---

Precluded (drone ship)	1
------------------------	---

During this period, the company achieved a 97% success rate in the 31 launch operations.

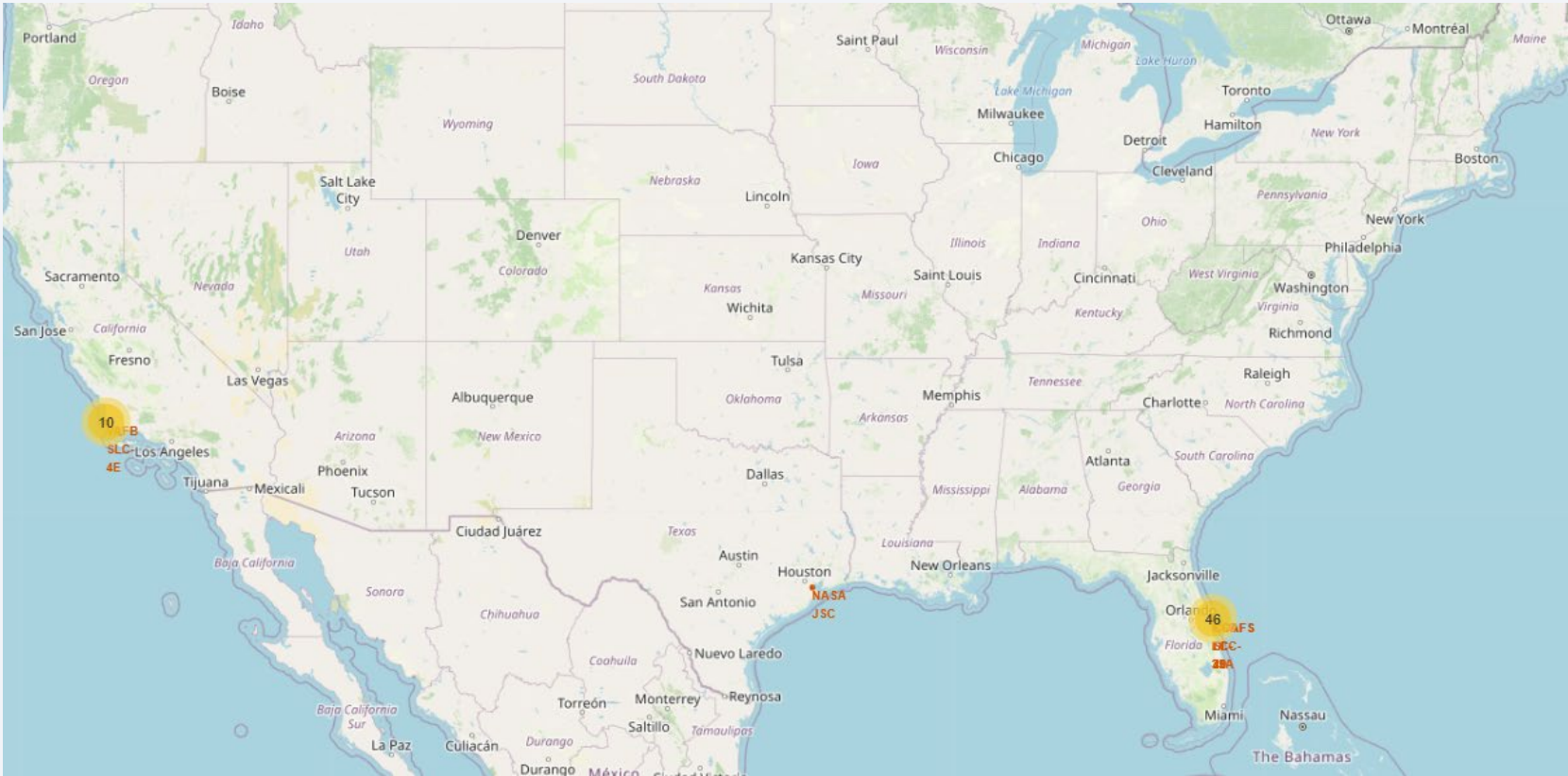
Excluding launches where there was no intention of recovery, the company achieved a 40% recovery rate for boosters.



Section 3

Launch Sites Proximities Analysis

The launch sites are located near the ocean, and the Cape Canaveral sites have the advantage of being closer to the equator



	Launch Site	Lat	Long
0	CCAFS LC-40	28.562302	-80.577356
1	CCAFS SLC-40	28.563197	-80.576820
2	KSC LC-39A	28.573255	-80.646895
3	VAFB SLC-4E	34.632834	-120.610745

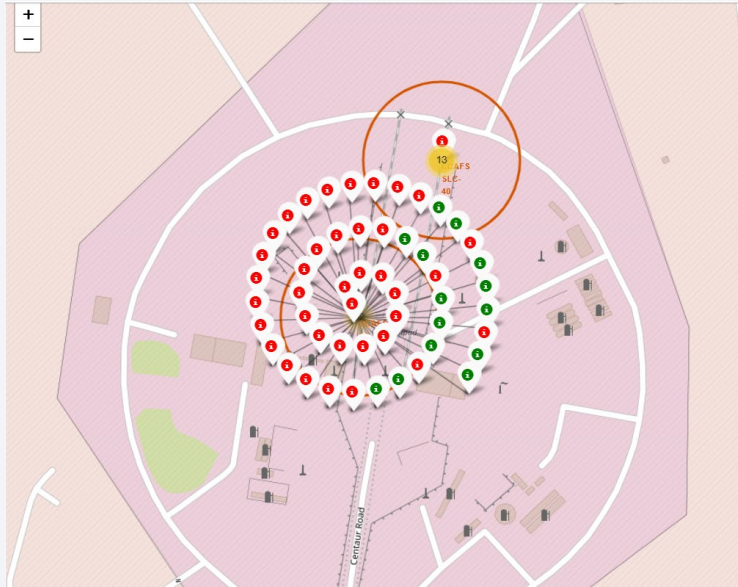
Launch centers are located at the latitudes closest to the equator possible within the American territory.

This is because rocket launch centers must seek locations near the equator to better utilize the momentum of the Earth's rotation. Launching near the equator saves fuel, increases payload, and maximizes the efficiency of the space mission.

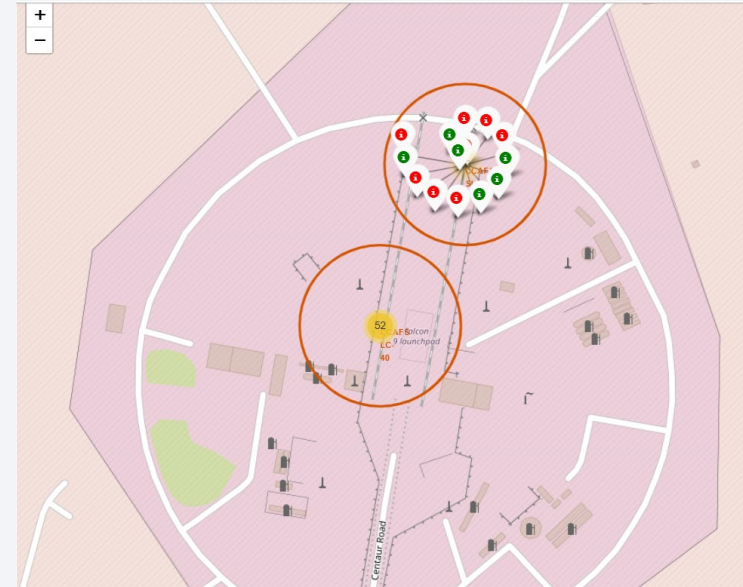
Probably Hawaii was not considered as a launch center location due to access logistics and the risk of seismicity.

The low success rate of the original site reflects the lessons learned, now applied to the redesigned site

LC-40 (Launch Complex 40):



SLC-40 (Space Launch Complex 40)



Historical and operational differences between LC-40 & SLC-40

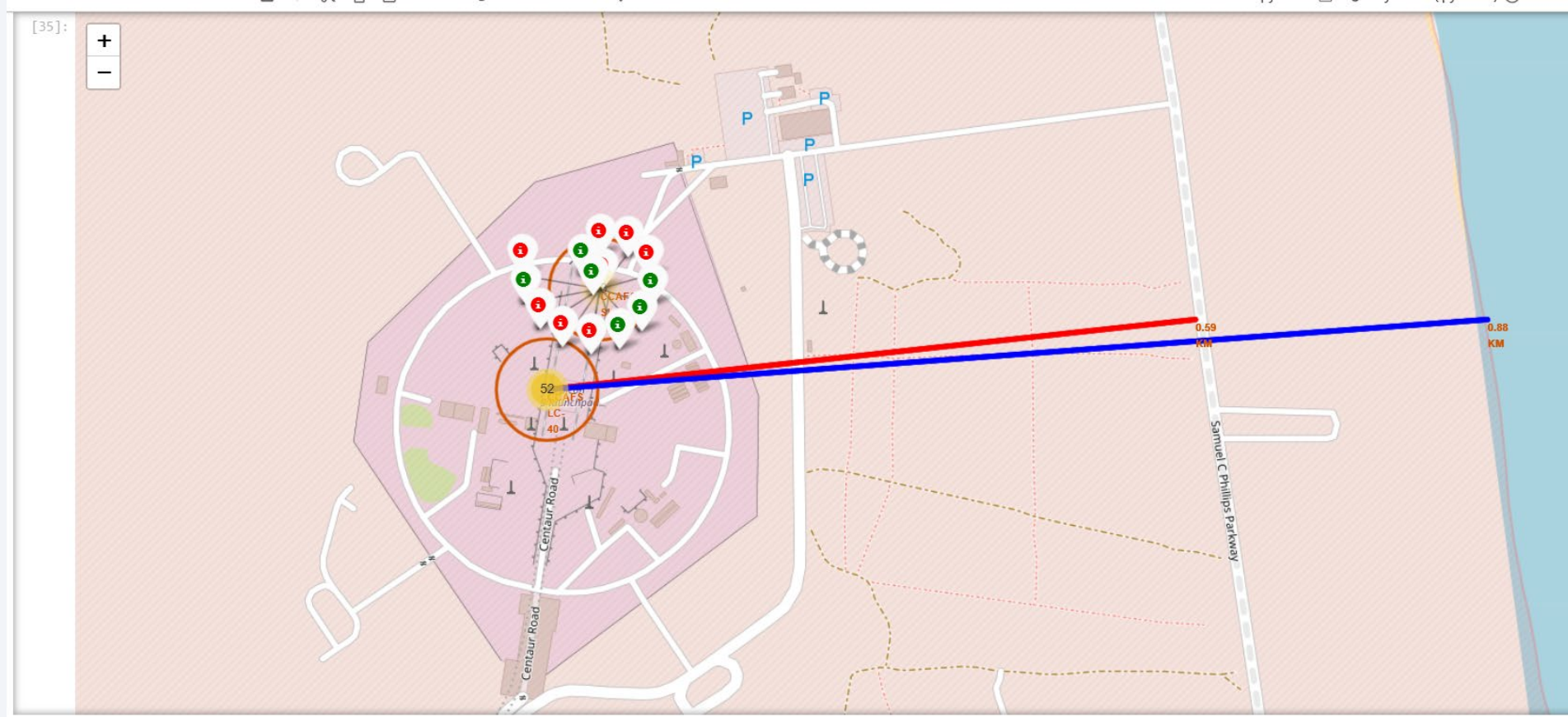
LC-40 (Launch Complex 40): This was the original name of the complex used by the US Air Force during the 1960s to 2000s for Titan III and Titan IV rockets.

SLC-40 (Space Launch Complex 40): This is the current name of the facility after being leased by SpaceX in 2007, rebuilt and upgraded to launch the Falcon 9 rocket.

Functional Difference: LC-40 was demolished/retrofitted to become SLC-40, which is now a SpaceX "workhorse" for satellite launches (Starlink, cargo missions and Dragon crew).

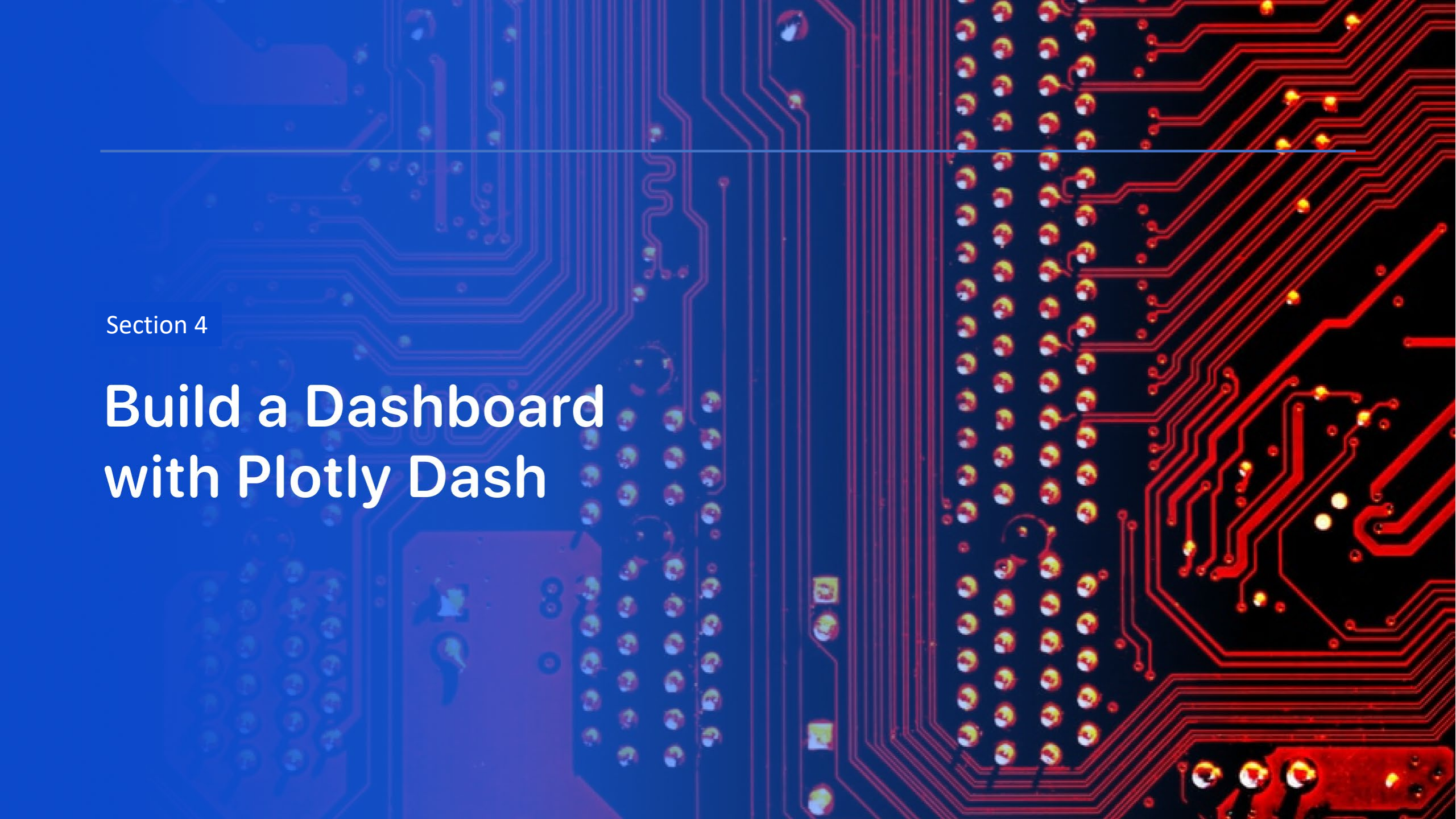
Summary: SLC-40 is the modern and upgraded version of the old LC-40, operated by SpaceX for the Falcon 9

The launch site is quite close to the highway and the coast



The launch site is 0.59 km from the highway and 0.98 km from the coastline

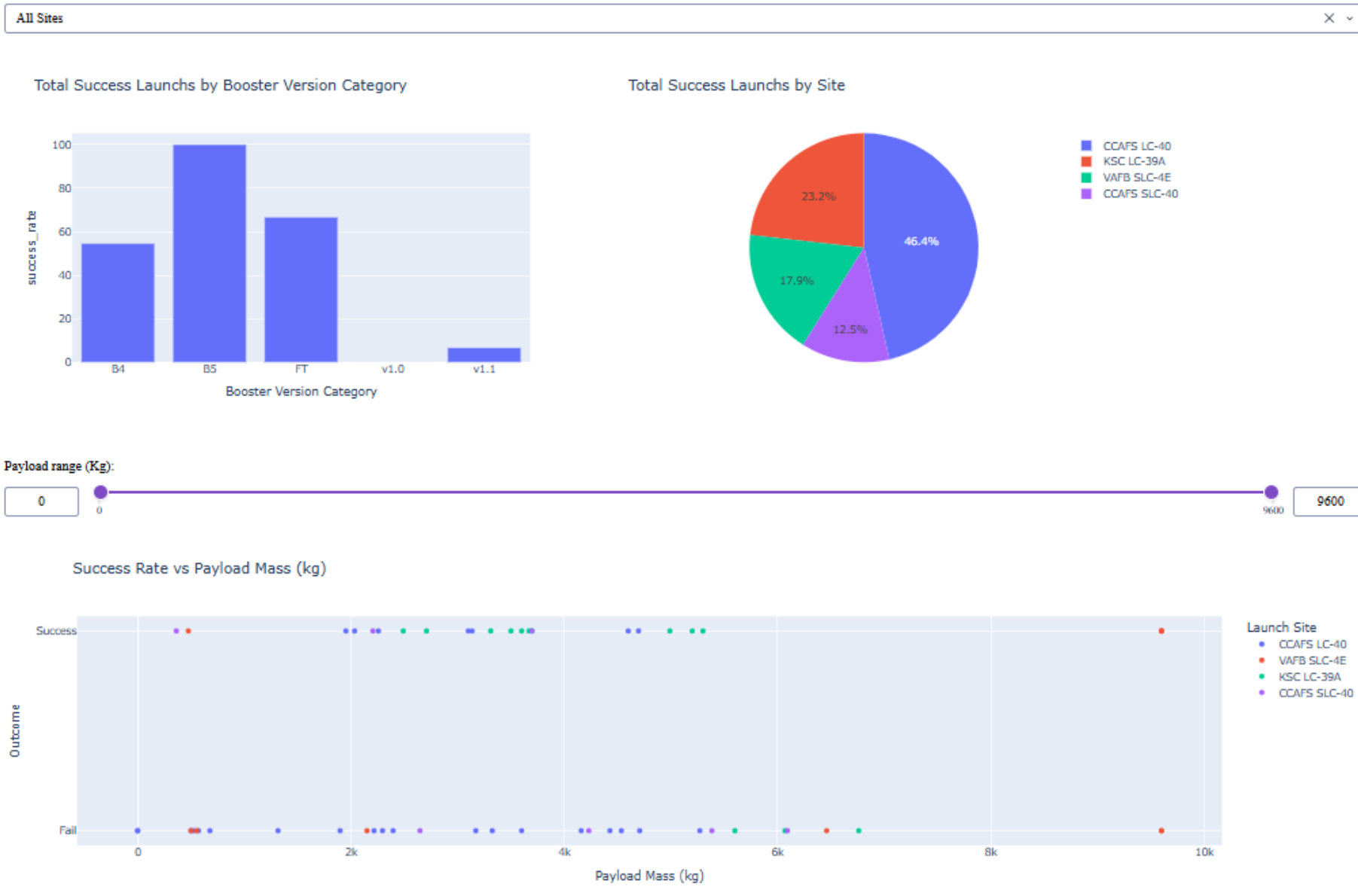
Proximity to the highway is an important factor in the logistics of the launch site operation



Section 4

Build a Dashboard with Plotly Dash

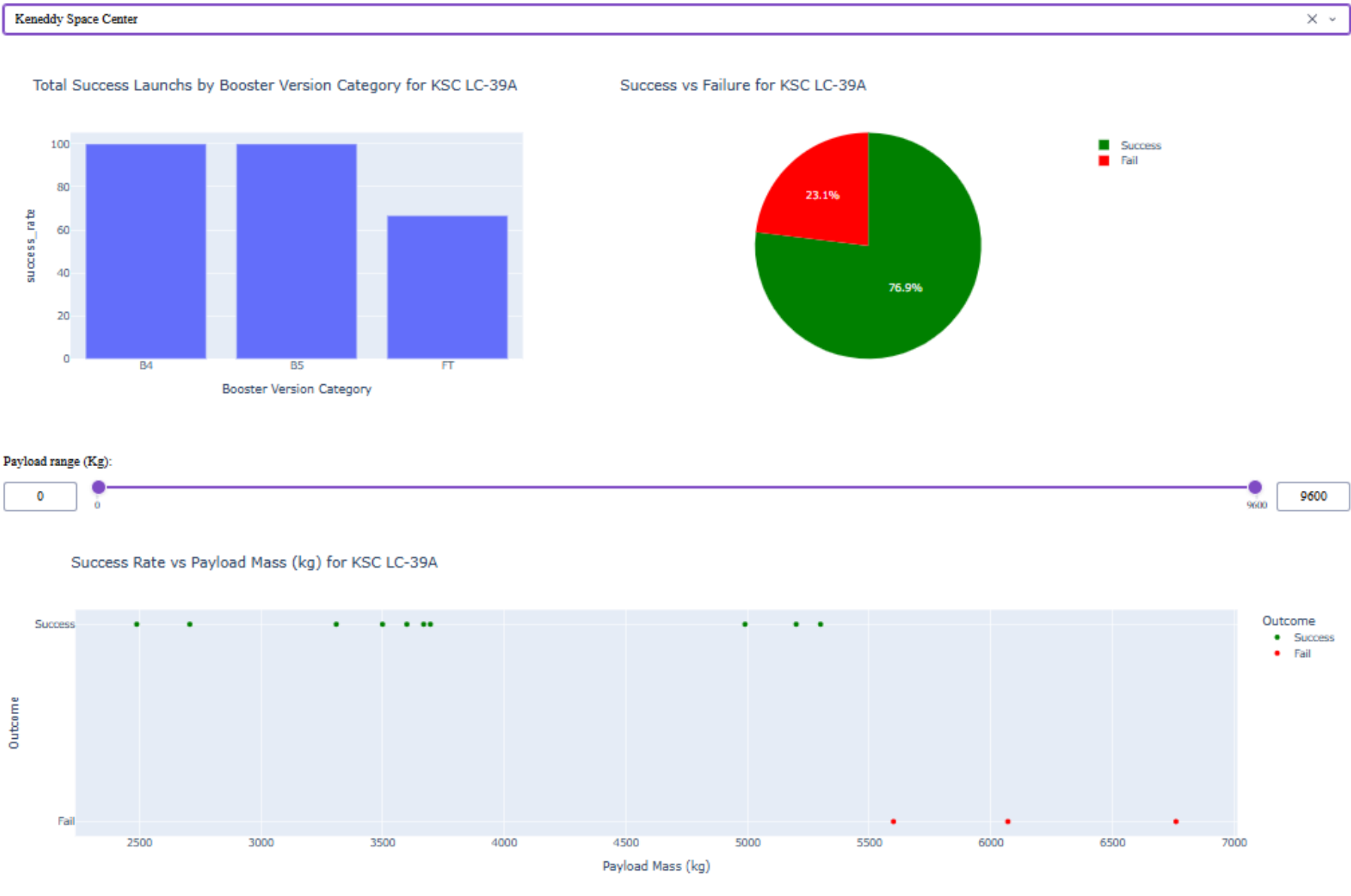
SpaceX Launch Records Dashboard



To obtain more information from the SpaceX dataset, a dashboard was created that displays it more easily than static graphs

- This dashboard allows you to view the success rate, payloads sent, and boosters used.
- Choosing the view for all sites allows you to understand their contributions to the overall success rate.
- You can choose the payload range for the overall view as well as for a specific site.

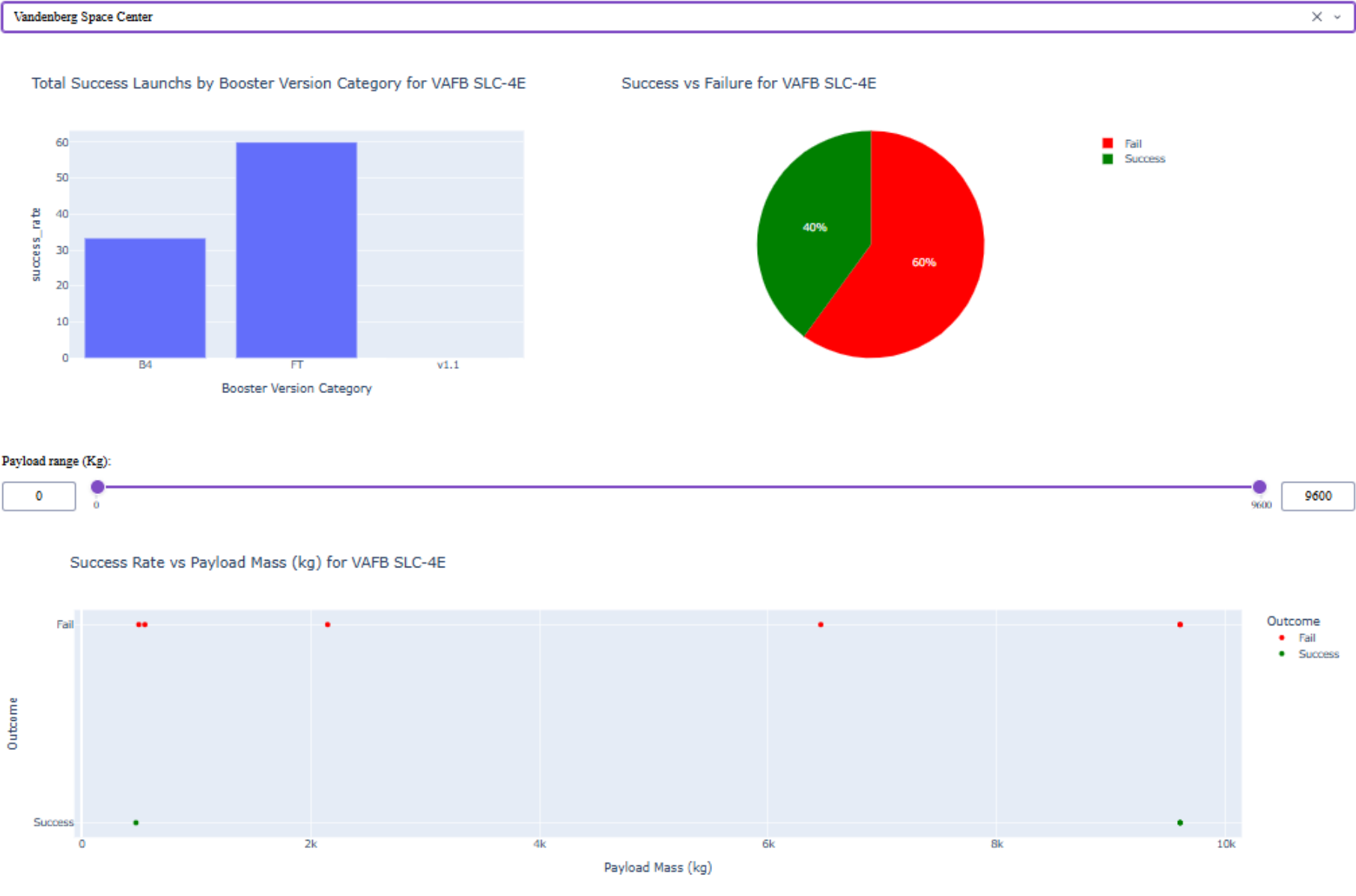
SpaceX Launch Records Dashboard



Kennedy Space Center

- The site reports a 76,9% success rate for shipments up to 5200kg;
- however, all shipments with loads exceeding this limit were unsuccessful

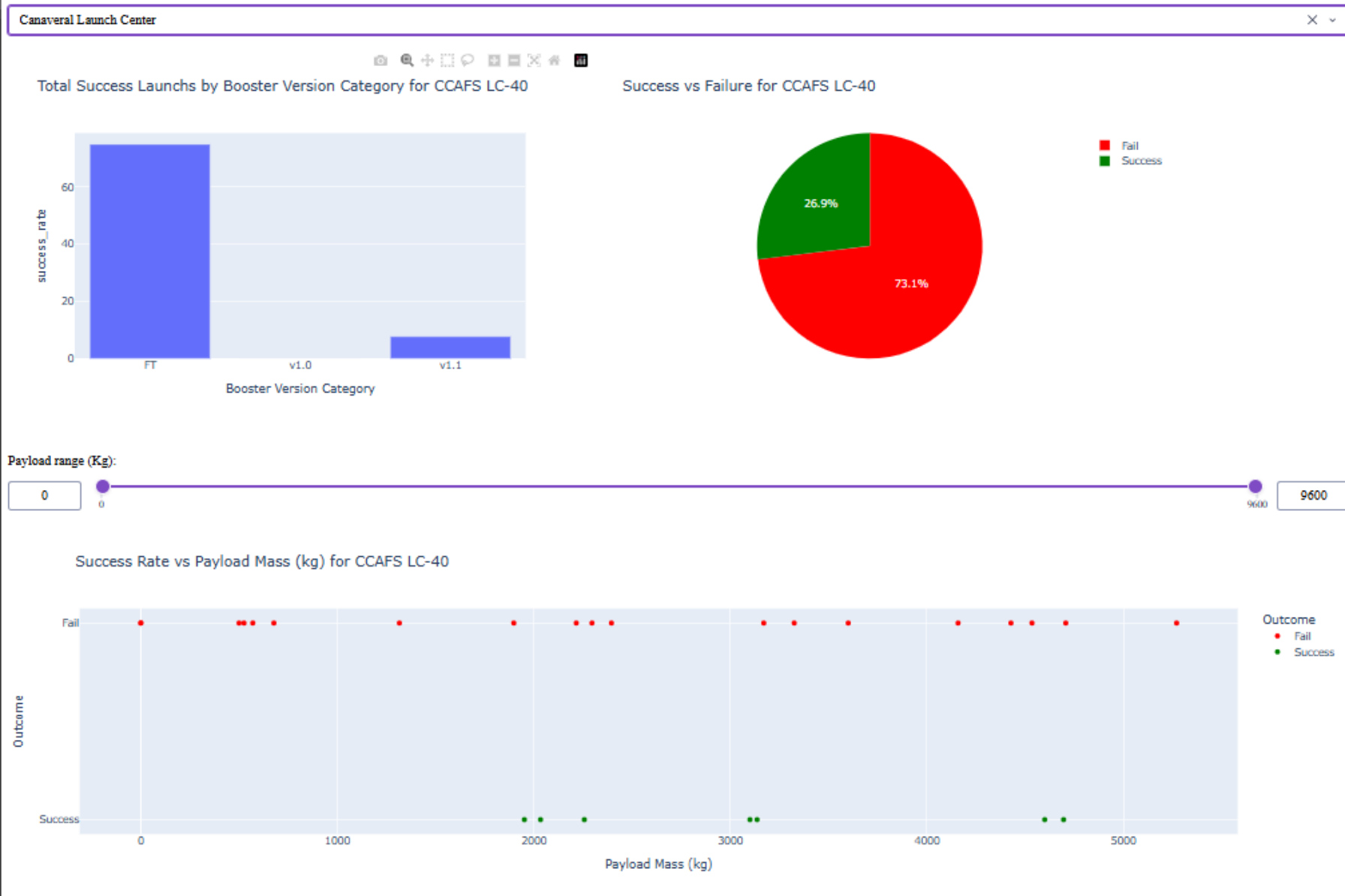
SpaceX Launch Records Dashboard



Vandenberg Space Center

- The site reports a 50% success rate for shipments of 10,000kg
- All other shipments below that weight were unsuccessful

SpaceX Launch Records Dashboard



Canaveral Launch Center

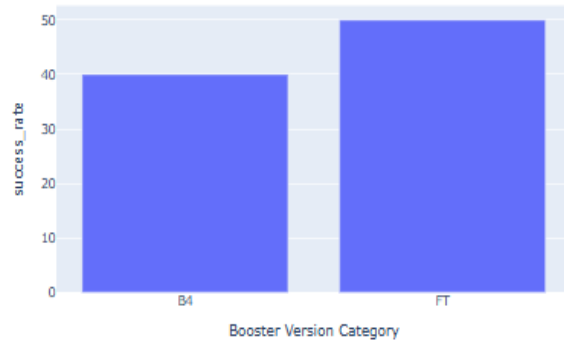
- The site boasts a 27% success rate for payloads between 2000 kg and 5000 kg.
- Several launches with payloads below this were unsuccessful.
- It's important to remember that this site has accumulated the learning from its first years of operation.

SpaceX Launch Records Dashboard

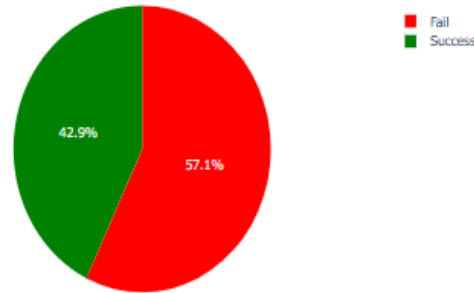
Canaveral Space Launch Center

X

Total Success Launches by Booster Version Category for CCAFS SLC-40



Success vs Failure for CCAFS SLC-40



- The site boasts a 43% success rate for payloads up to 3800 kg
- Several launches with payloads upper this were unsuccessful.

Payload range (Kg):

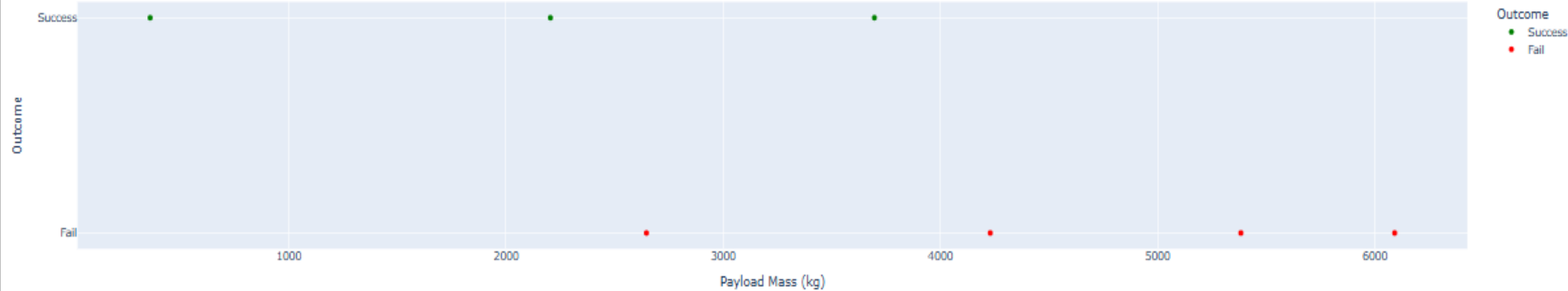
0

0

9600

9600

Success Rate vs Payload Mass (kg) for CCAFS SLC-40



Section 5

Predictive Analysis (Classification)

The resulting database underwent a process of isolating the target variable from the predictor variables, followed by normalization or standardization (-score) of the features

	FlightNumber	PayloadMass	Flights	Block	ReusedCount	Orbit_ES-L1	Orbit_GEO	Orbit_GTO	Orbit_HEO	Orbit_ISS	...	Serial_B1058	Serial_B1059	Serial_B1060	Serial_B1062	GridFins_False	GridFins_True	Reused_
0	1.0	6104.959412	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	1.0	0.0	
1	2.0	525.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	1.0	0.0	
2	3.0	677.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0	...	0.0	0.0	0.0	0.0	1.0	0.0	
3	4.0	500.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	1.0	0.0	
4	5.0	3170.000000	1.0	1.0	0.0	0.0	0.0	1.0	0.0	0.0	...	0.0	0.0	0.0	0.0	1.0	0.0	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
85	86.0	15400.000000	2.0	5.0	2.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	1.0	0.0	0.0	1.0	
86	87.0	15400.000000	3.0	5.0	2.0	0.0	0.0	0.0	0.0	0.0	...	1.0	0.0	0.0	0.0	0.0	1.0	
87	88.0	15400.000000	6.0	5.0	5.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	1.0	
88	89.0	15400.000000	3.0	5.0	2.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	1.0	0.0	0.0	1.0	
89	90.0	3681.000000	1.0	5.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	1.0	0.0	1.0	

90 rows × 83 columns

- The process of separating and standardizing data in machine learning involves isolating the target variable () from the predictor variables (features,), followed by normalization or standardization (-score) of the features.
- This ensures that variables with larger scales do not dominate the model, improving the convergence of distance- or gradient-based algorithms

The next step is to divide the data into training and testing databases

```
X_train, X_test, Y_train, Y_test  
:  
: X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=2)  
:  
: Y_test.shape  
:  
: (18,)  
:  
: Y_train.shape  
:  
: (72,)
```

we can see we only have 18 test samples.

- We split the data into training and testing data using the function `train_test_split`.
- The training data is divided into validation data, a second set used for training data;
- Then the models are trained and hyperparameters are selected using the function `GridSearchCV`.

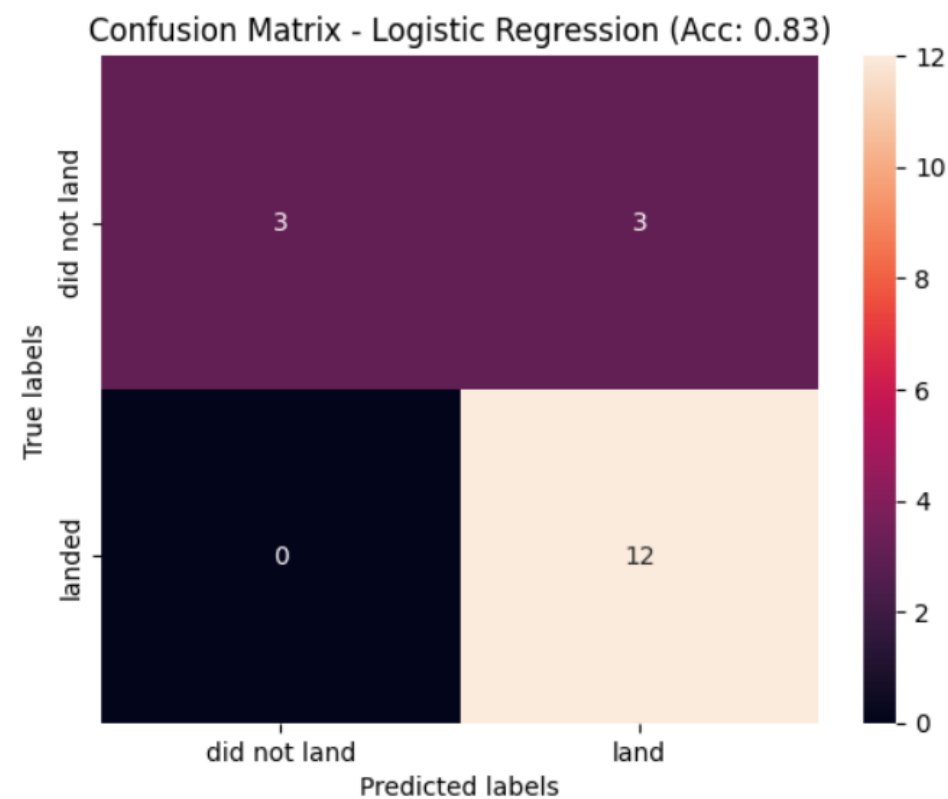
To optimize performance, the data underwent a hyperparameter tuning process for four distinct models: SVM, KNN, Classification Trees, and Logistic Regression

Key qualities and strengths of the four classification models:

- **Support Vector Machine (SVM):** Highly effective in high-dimensional spaces, making it ideal for text and image classification. It is robust against overfitting, particularly in high-dimensional scenarios, and versatile due to various kernel functions that handle non-linear data efficiently.
- **K-Nearest Neighbors (KNN):** Simple to understand and implement, making it great for quick prototyping. As a lazy learner, it requires no training phase and is non-parametric, meaning it makes no assumptions about data distribution, making it flexible for complex, non-linear datasets.
- **Classification Trees:** Highly interpretable and transparent, allowing for easy visualization of decision-making. They require little data preprocessing (no scaling needed) and naturally handle both numerical and categorical data, as well as complex feature interactions.
- **Logistic Regression:** Efficient, fast, and highly interpretable, providing probabilistic insights rather than just binary labels. It works very well for linearly separable data and is less prone to overfitting in low-dimensional datasets.

Source: Wikipedia and IA

Logistic Regression cassification metrics



Tuned hpyerparameters :
(best parameters) {'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}

Accuracy : 0.8464285714285713

Logistic Regression Accuracy (score method): 0.8333333333333334

Mean Absolute Error (MAE): 0.1667

Mean Squared Error (MSE): 0.1667

Root Mean Squared Error (RMSE): 0.4082

R2 Score (r2): 0.2500

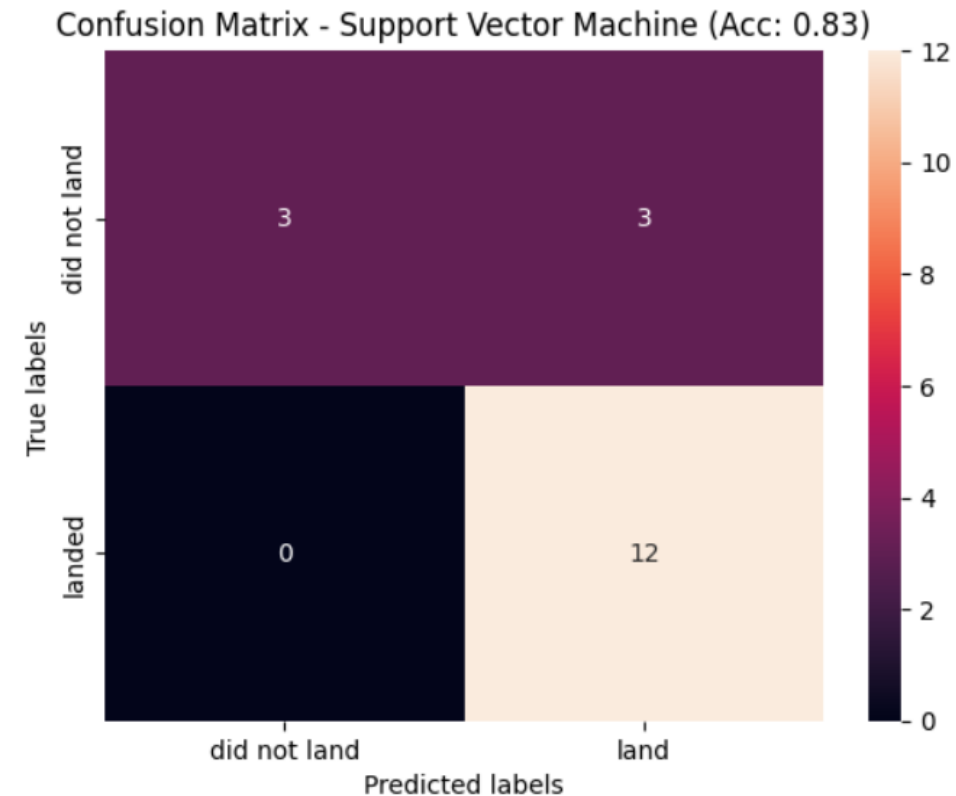
precision	recall	f1-score	support	
Did not Land	1.00	0.50	0.67	6
Landed	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18

Overview:

True Postive - 12 (True label is landed, Predicted label is also landed)

False Postive - 3 (True label is not landed, Predicted label is landed)

Support Vector Machine cassification metrics



Tuned hpyerparameters :
(best parameters) {'C': np.float64(1.0), 'gamma': np.float64(0.03162277660168379), 'kernel': 'sigmoid'}

accuracy : 0.8482142857142856

Support Vector Machine Accuracy (score method): 0.8333333333333334

Mean Absolute Error (MAE): 0.1667

Mean Squared Error (MSE): 0.1667

Root Mean Squared Error (RMSE): 0.4082

R2 Score (r2): 0.2500

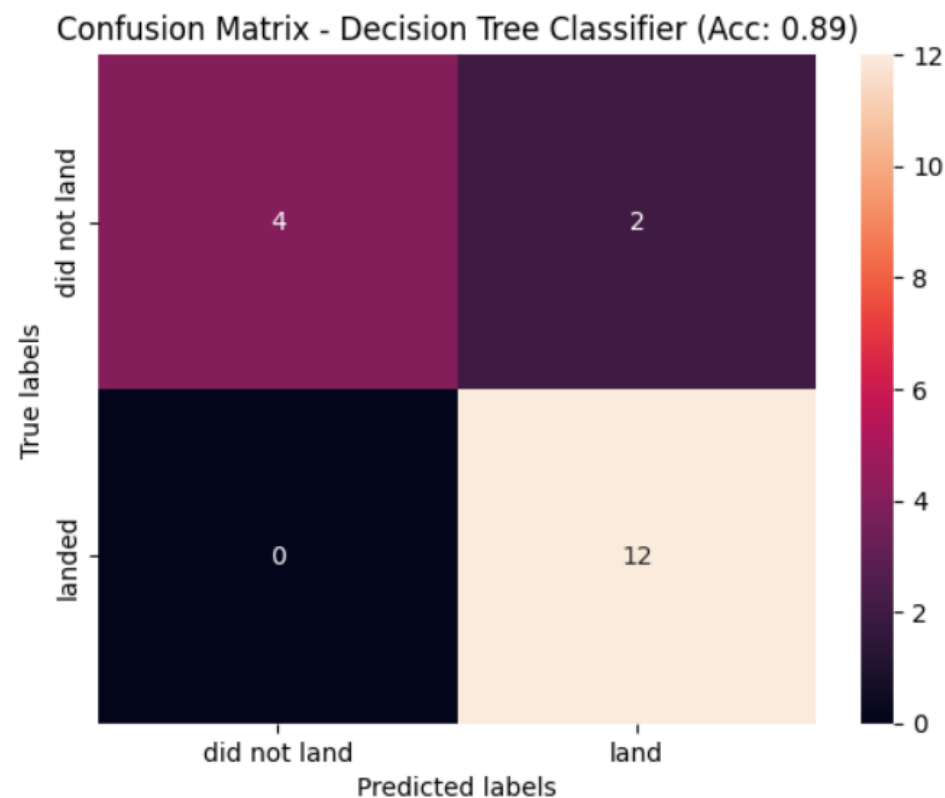
precision	recall	f1-score	support	
Did not Land	1.00	0.50	0.67	6
Landed	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18

Overview:

True Postive - 12 (True label is landed, Predicted label is also landed)

False Postive - 3 (True label is not landed, Predicted label is landed)

Decision Tree Classifier classification metrics



Tuned hpyerparameters :
(best parameters) {'algorithm': 'auto', 'n_neighbors': 10, 'p': 1}
accuracy : 0.8482142857142858

Decision Tree Classifier Accuracy (score method): 0.8333333333333334

Mean Absolute Error (MAE): 0.1667
Mean Squared Error (MSE): 0.1667
Root Mean Squared Error (RMSE): 0.4082
R2 Score (r2): 0.2500

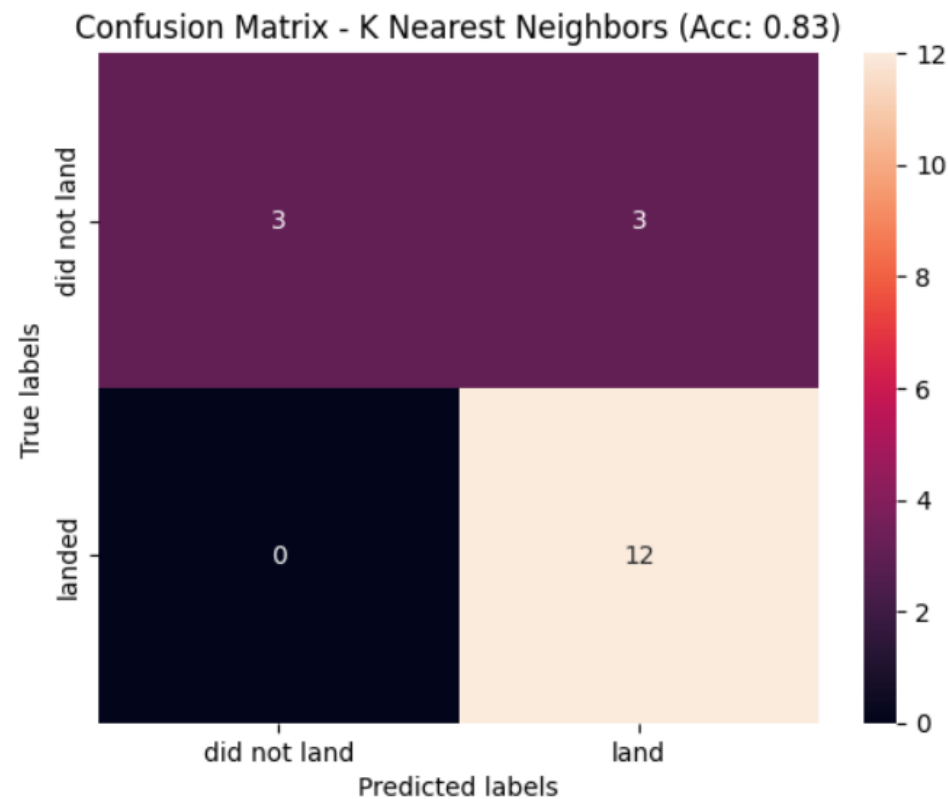
precision	recall	f1-score	support
Did not Land	1.00	0.50	0.676
Landed	0.80	1.00	0.8912
accuracy			0.8318
macro avg	0.90	0.75	0.7818
weighted avg	0.87	0.83	0.8118

Overview:

True Postive - 12 (True label is landed, Predicted label is also landed)

False Postive – 2 (True label is not landed, Predicted label is landed)

K Nearest Neighbors classifier classification metrics



Tuned hpyerparameters :
(best parameters) {'algorithm': 'auto', 'n_neighbors': 10, 'p': 1}
accuracy : 0.8482142857142858

Decision Tree Classifier Accuracy (score method): 0.8333333333333334

Mean Absolute Error (MAE): 0.1667
Mean Squared Error (MSE): 0.1667
Root Mean Squared Error (RMSE): 0.4082
R2 Score (r2): 0.2500

precision	recall	f1-score	support	
Did not Land	1.00	0.50	0.67	6
Landed	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18

Overview:

True Postive - 12 (True label is landed, Predicted label is also landed)

False Postive – 2 (True label is not landed, Predicted label is landed)

Model comparison results summary

1. The comparison of the models using `best_score_` from cross-validation shows that Decision Tree had the highest accuracy (85.7%).
2. The SVM and KNN models were practically the same (~84.8%), and Logistic Regression had the lowest value, although the difference is small (84.6%).
3. The Decision Tree appears to be the best model according to cross-validation, but the difference is small (1%).
4. This suggests that all models have similar performance. The test set used is relatively small (18 samples), which may limit the reliability of the performance metrics. Small variations in predictions can cause significant changes in accuracy.

Section 6

Proposed formulas for determining costs

Appendix: Determining the cost

1.) Using only recovery success rate

Example of calculation reasoning for determining the cost by applying the first-stage recovery success rate

```
recovery_rate = np.mean(Y)
print(f"Historical Recovery Rate : {recovery_rate:.4f}")
```

Historical Recovery Rate : 0.6667

```
cost_new = 64      # US$ millions - Hypothetical values
```

```
cost_reused = 42   # US$ millions - Hypothetical values for cost of reuse, cost if recovery attempt fails, indirect cost of attempting landing (less payload)
```

```
expected_cost = cost_new * (1 - recovery_rate) + cost_reused * recovery_rate
```

```
print(f"Expected cost (US$ millions) : {expected_cost:.2f}")
```

Expected cost (US\$ millions) : 49.33

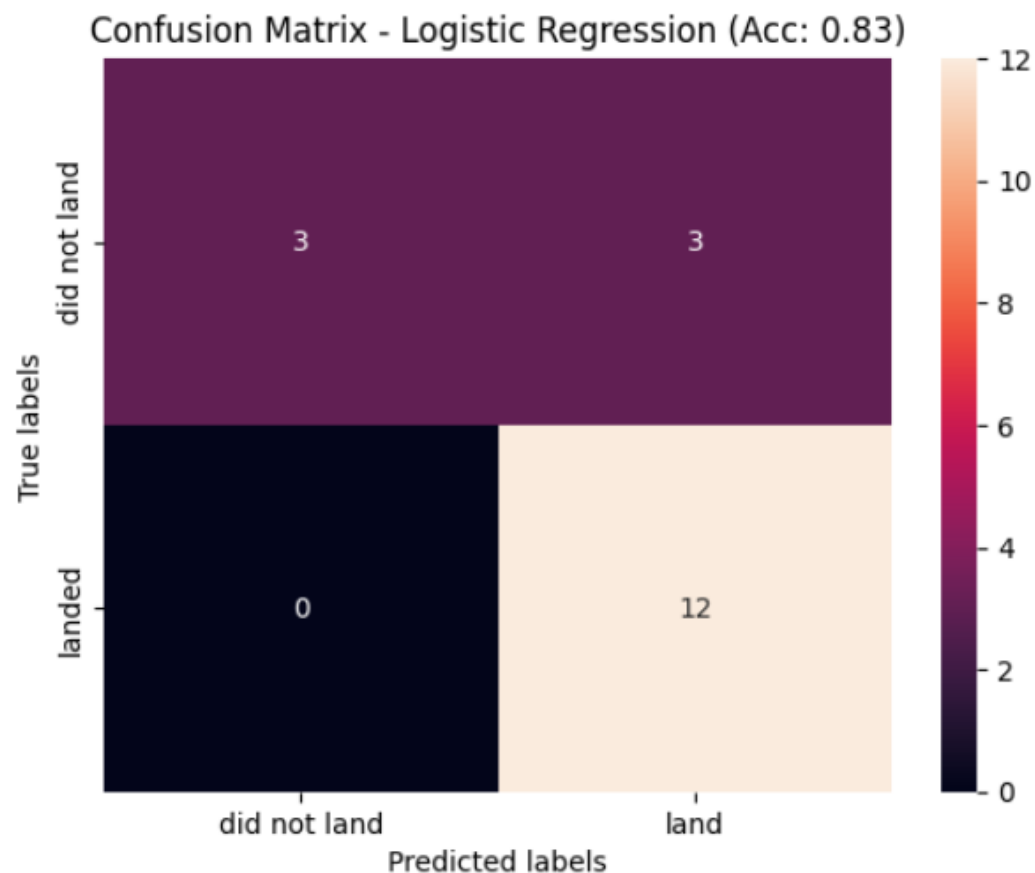
This preliminary draft was developed based on consultations with Wikipedia and IA

Appendix: Determining the cost

2.) Calibrate the historical average

Using confusion matrix metrics to calibrate the historical average

```
[83]: plot_confusion_matrix(Y_test,yhat, 'Logistic Regression', accuracy )
```



TP = 12

FN = 0

TN = 3

FP = 3

$TPR = TP / (TP + FN)$ # hit rate on recoverable items

$FPR = FP / (FP + TN)$ # miss rate on non-recoverable items

$recovery_rate_model = recovery_rate * TPR + (1 - recovery_rate) * FPR$

$expected_cost = cost_new * (1 - recovery_rate_model) + cost_reused * recovery_rate_model$
`print(f"Expected cost (US$ millions) : {expected_cost:.2f}")`

Expected cost (US\$ millions) : 45.67

Appendix: Determining the cost

3.) Reuse or Discard decision

Assessing the probability of successful recovery to decide: Reuse (attempt landing) or Discard (use everything for performance).

```
cost_new = 64          # cost of a new booster → US$ 64M
cost_reuse = 10         # cost of reuse (refurbishment) → US$ 10M
cost_fail = 64          # cost if recovery attempt fails → US$ 64M
cost_fuel_penalty = 5   # indirect cost of attempting landing (less payload) → US$ 5M
```

```
reused_estrategy = cost_reuse * recovery_rate + cost_fail * (1 - recovery_rate) + cost_fuel_penalty
discard_estrategy = cost_new
```

```
print(f"Reuse cost (US$ millions): {reused_estrategy:.2f}")
print(f"Discard cost (US$ millions): {discard_estrategy:.2f}")
```

```
Reuse cost (US$ millions): 33.00
Discard cost (US$ millions): 64.00
```

```
if reused_estrategy < discard_estrategy:
    print(f"Reuse cost (US$ millions): {reused_estrategy:.2f}")
else:
    print(f"Discard cost (US$ millions): {discard_estrategy:.2f}");
```

```
Reuse cost (US$ millions): 33.00
```

Conclusions

Although different algorithms were tested, all produced identical results on the test set, suggesting that the problem may be relatively simple or that the size of the dataset limits the ability to distinguish performance between models.

The recall for the Did not Land class is relatively low (0.50), indicating that the model fails to correctly identify half of the cases in which the rocket does not land. Predicting that a rocket will land when it will not can be very costly.

For this application, this could represent a significant risk and suggests the need for improvements to the model or data balancing.

Appendix

- This appendix was compiled primarily from the procedures and explanations of the labs conducted, as well as research from other sources, and served as a sandbox for the main presentation

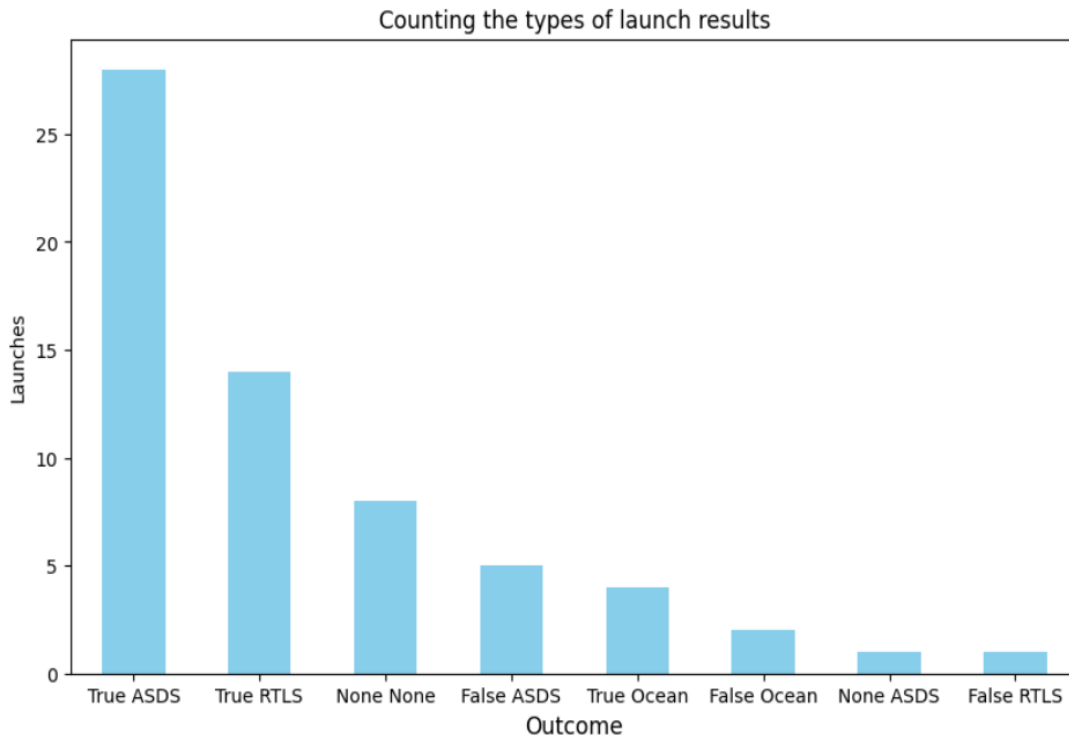
Views



<https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBMDeveloperSkillsNetwork-DS0701EN-SkillsNetwork/api/Images/crash.gif>

https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBMDeveloperSkillsNetwork-DS0701EN-SkillsNetwork/api/Images/landing_1.gif

Successful recovery from the first stage can occur on a drone ship, on the ground pad, or in a specific region of the oceans



True Ocean means the mission outcome was successfully landed to a specific region of the ocean while

False Ocean means the mission outcome was unsuccessfully landed to a specific region of the ocean.

True RTLS means the mission outcome was successfully landed to a ground pad

False RTLS means the mission outcome was unsuccessfully landed to a ground pad.

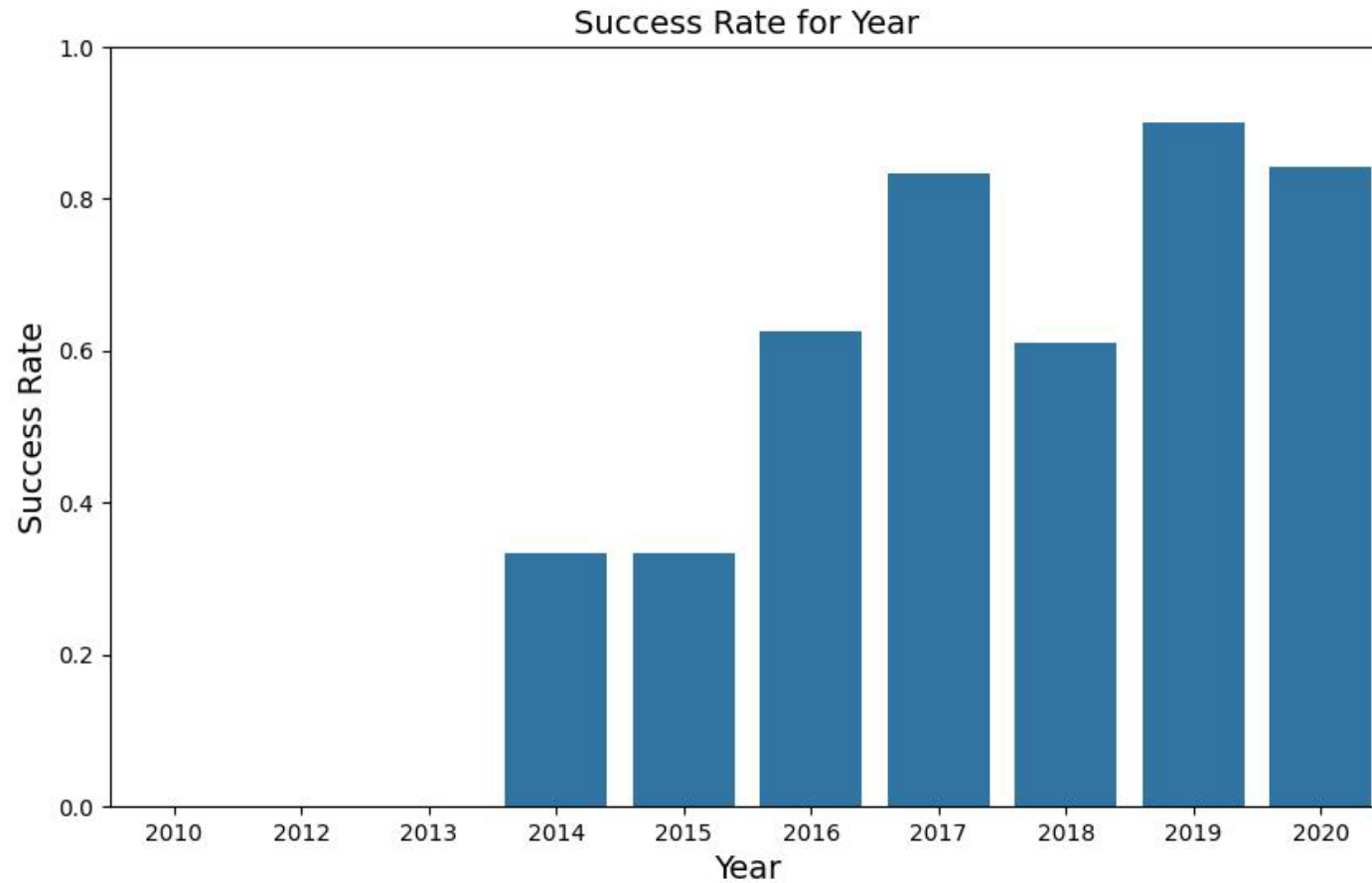
True ASDS means the mission outcome was successfully landed to a drone ship

False ASDS means the mission outcome was unsuccessfully landed to a drone ship.

None ASDS and **None None** these represent a failure to land.

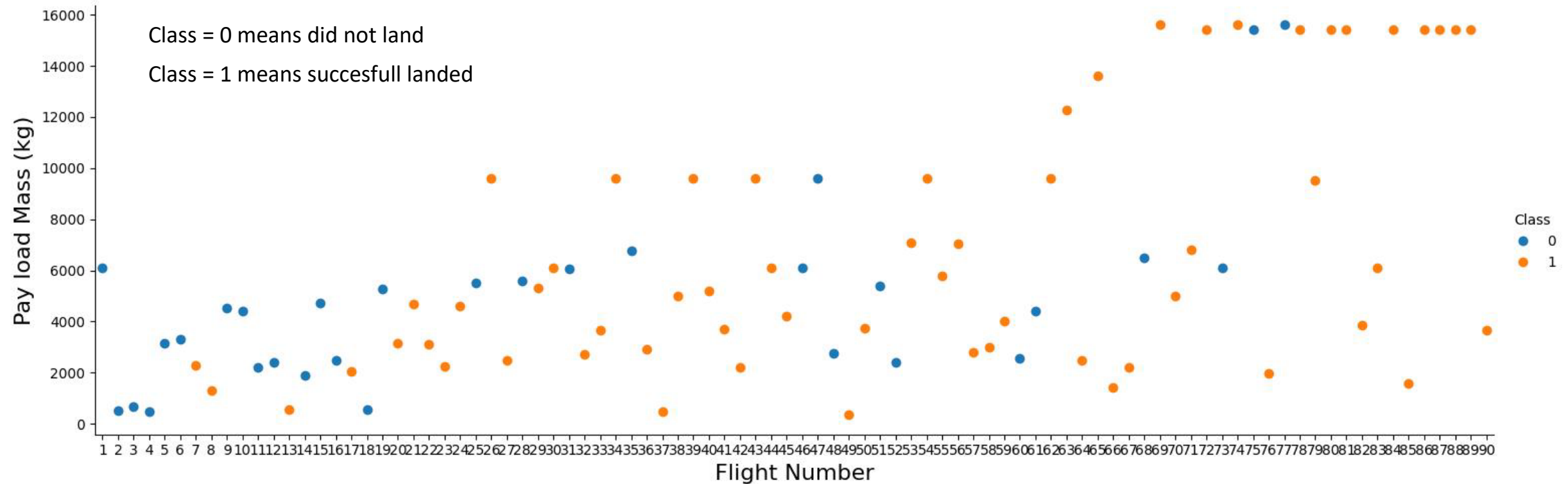
The success indicator label is created from the selection of successful outcome types

The annual trend of successful launches has been increasing



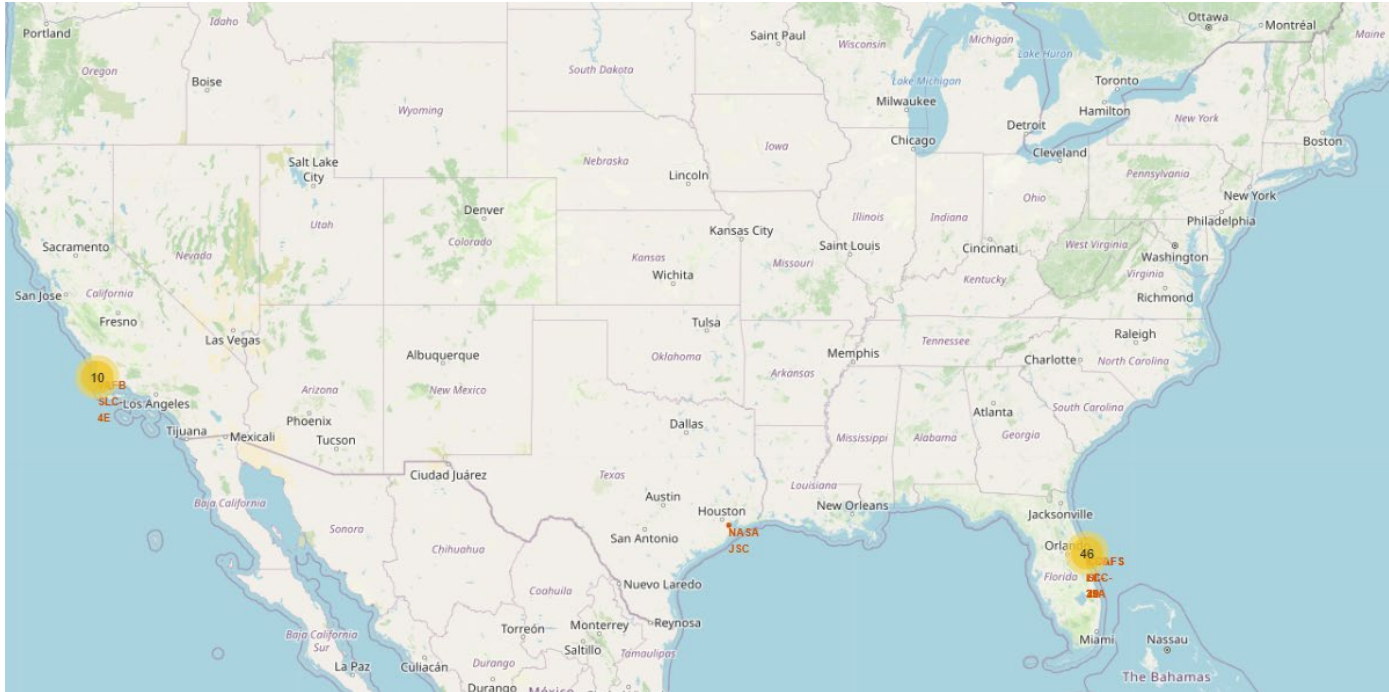
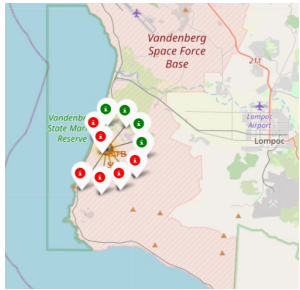
- The success rate since 2013 kept increasing till 2017 (stable in 2014) and after 2015 it started increasing

The sequence of flight numbers related to payload mass shows how the success rate in first-stage recovery has progressed

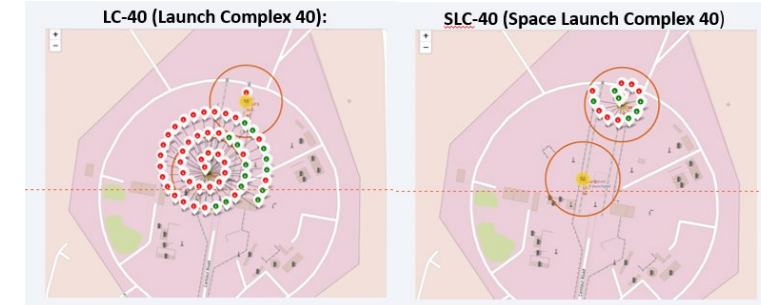


These launches were carried out from three launch sites, two in Florida and one in California

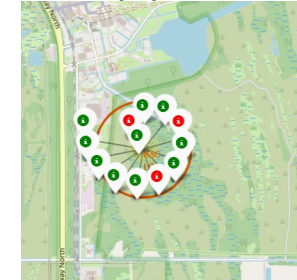
Vandenberg Space Force Base



Cape Canaveral



Kennedy Space Center

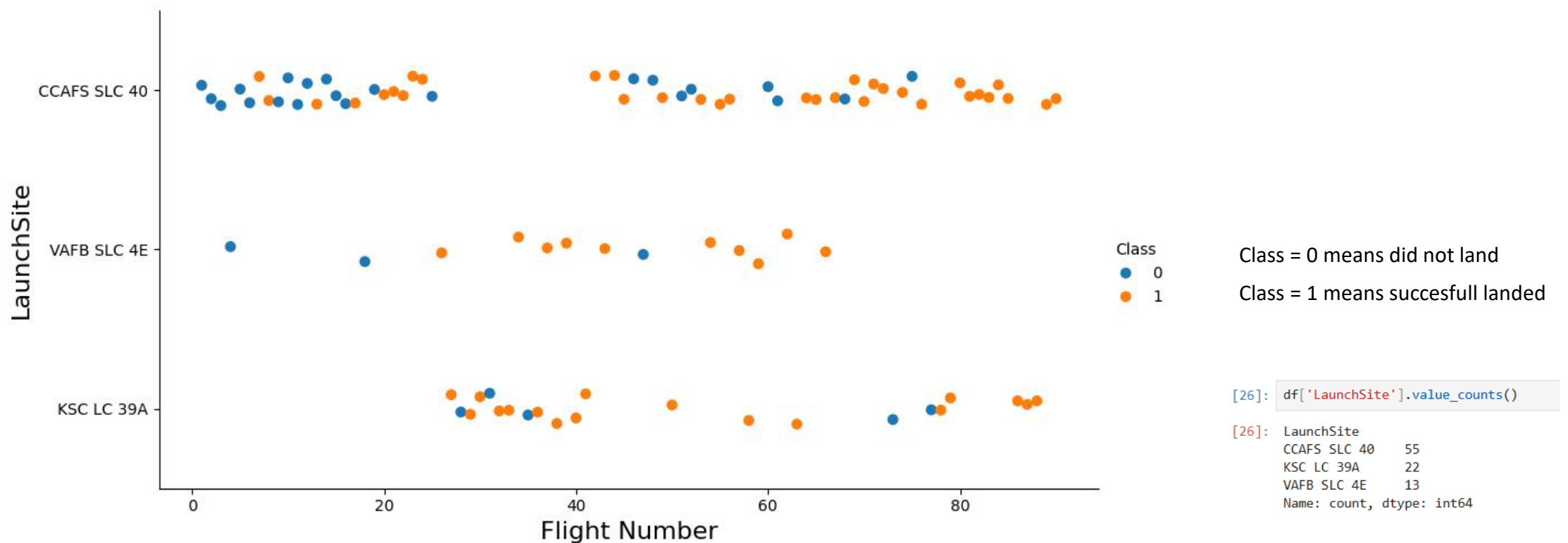


Launches began at the **Cape Canaveral LC-40 (Launch Complex 40)** launch site, which carried out almost all of SpaceX's first 26 launches

In the period between the 26th and 40th launches, the CCAFS launch site did not carry out any launches. Launches were carried out at the neighboring **Kennedy Space Center (KSC)** launch site, with 11 launches, 8 of which were successful, and at the **Vandenberg Space Force Base (VALFB)** launch site, which carried out 5 successful launches.

From the 40th launch onwards the **Cape Canaveral SLC-40 (Space Launch Complex 40)** Launch Site resumed operations with 33 launches.

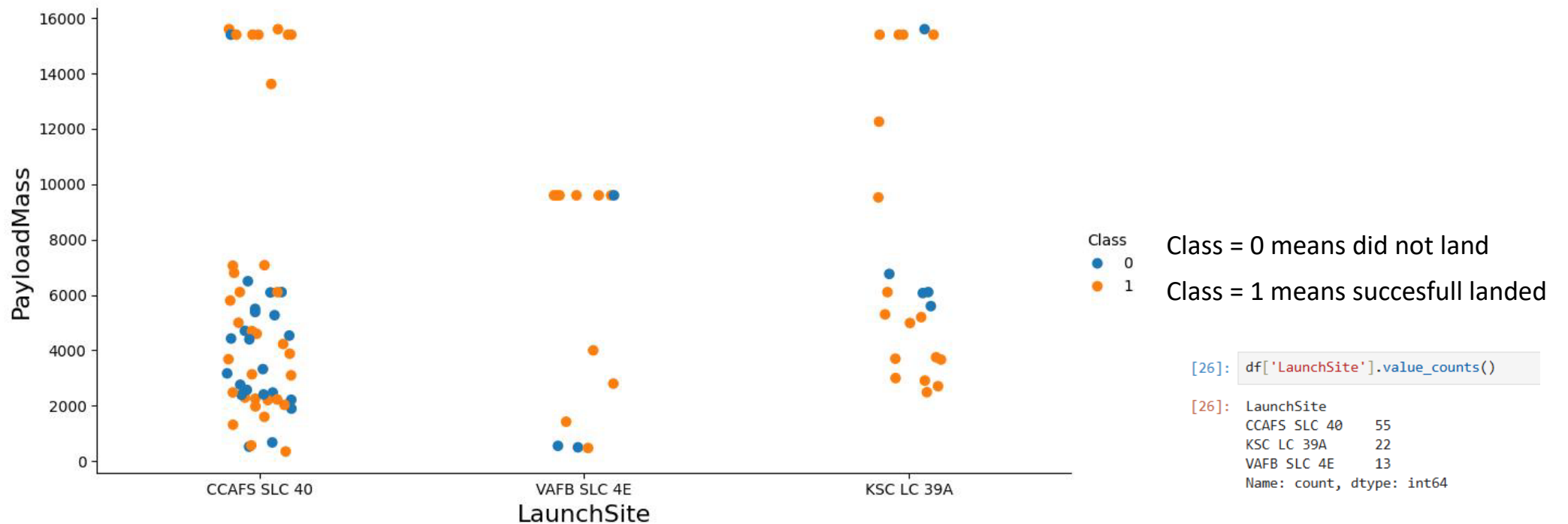
Ninety launches were conducted between June 2010 and May 2020, with a first-stage recovery success rate of 67%.



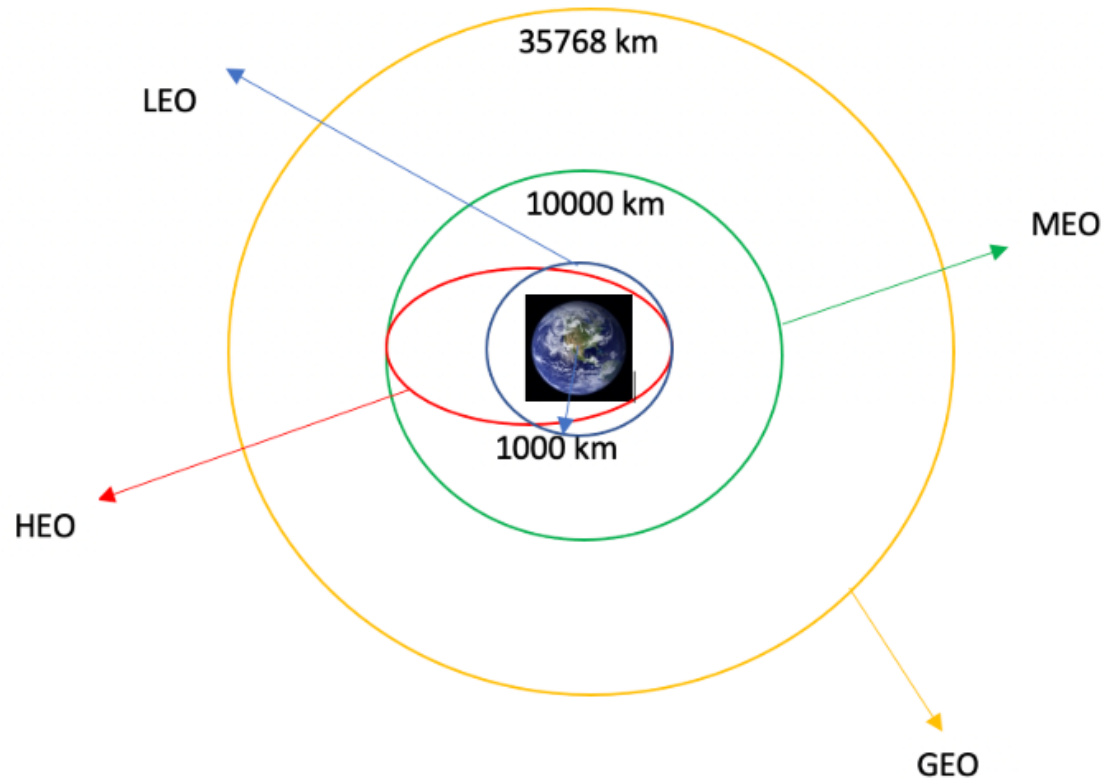
The launch success rate may depend on many factors such as payload mass, orbit type, and so on. It may also depend on the location and proximities of a launch site, i.e., the initial position of rocket trajectories.

Finding an optimal location for building a launch site certainly involves many factors and hopefully we could discover some of the factors by analyzing the existing launch site locations.

The VAFB-SLC launchsite there are no rockets launched for heavypayload mass (greater than 10000)



There are various orbits with different purposes that can be reached by launches



Each launch aims to an dedicated orbit, and here are some common orbit types

LEO: Low Earth orbit (LEO) is an Earth-centred orbit with an altitude of 2,000 km (1,200 mi) or less (approximately one-third of the radius of Earth), [1] or with at least 11.25 periods per day (an orbital period of 128 minutes or less) and an eccentricity less than 0.25. [2] Most of the manmade objects in outer space are in LEO [1].

VLEO: Very Low Earth Orbits (VLEO) can be defined as the orbits with a mean altitude below 450 km. Operating in these orbits can provide a number of benefits to Earth observation spacecraft as the spacecraft operates closer to the observation [2].

GTO (Geostationary Transfer Orbit): A geostationary transfer orbit is an elliptical Earth orbit used to transfer satellites from Low Earth orbit (LEO) to geostationary orbit (GEO). In a GTO, the perigee (closest point to Earth) is much lower than GEO altitude, while the apogee (farthest point) reaches approximately 22,236 miles (35,786 kilometers) above Earth's equator — the altitude of a geostationary orbit. Satellites in GTO use onboard propulsion to circularize their orbit at GEO altitude, where they can provide services such as weather monitoring, communications, and surveillance. [3].

SSO (or SO): It is a Sun-synchronous orbit also called a heliosynchronous orbit is a nearly polar orbit around a planet, in which the satellite passes over any given point of the planet's surface at the same local mean solar time [4].

ES-L1: At the Lagrange points the gravitational forces of the two large bodies cancel out in such a way that a small object placed in orbit there is in equilibrium relative to the center of mass of the large bodies. L1 is one such point between the sun and the earth [5].

HEO A highly elliptical orbit, is an elliptic orbit with high eccentricity, usually referring to one around Earth [6].

ISS A modular space station (habitable artificial satellite) in low Earth orbit. It is a multinational collaborative project between five participating space agencies: NASA (United States), Roscosmos (Russia), JAXA (Japan), ESA (Europe), and CSA (Canada) [7].

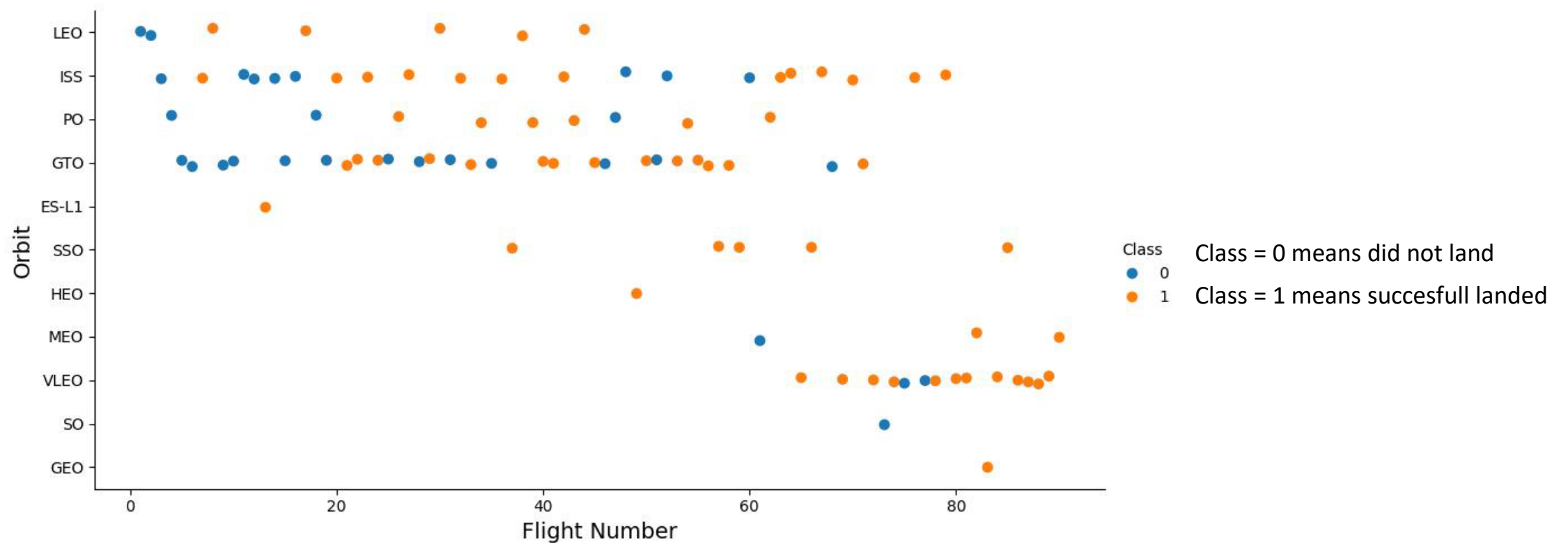
MEO Geocentric orbits ranging in altitude from 2,000 km (1,200 mi) to just below geosynchronous orbit at 35,786 kilometers (22,236 mi). Also known as an intermediate circular orbit. These are "most commonly at 20,200 kilometers (12,600 mi), or 20,650 kilometers (12,830 mi), with an orbital period of 12 hours [8].

HEO Geocentric orbits above the altitude of geosynchronous orbit (35,786 km or 22,236 mi) [9].

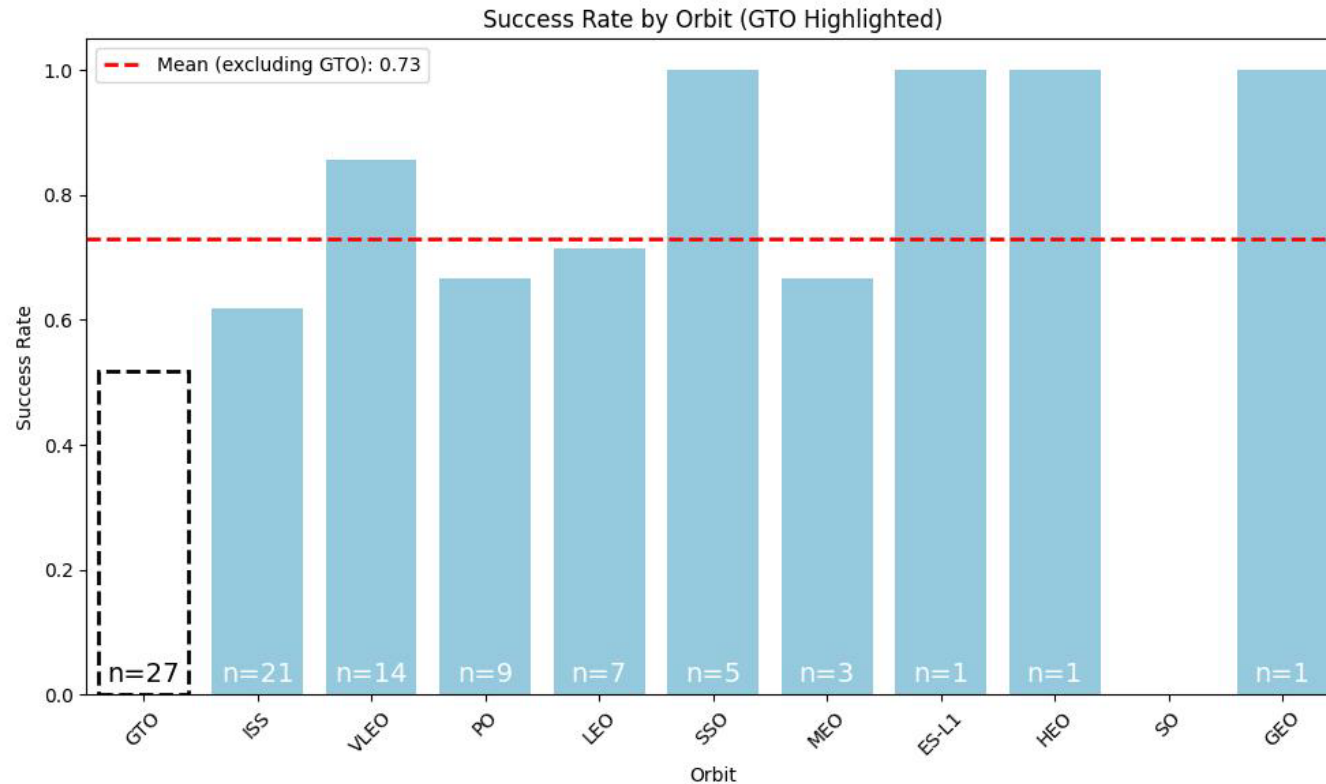
GEO It is a circular geosynchronous orbit 35,786 kilometres (22,236 miles) above Earth's equator and following the direction of Earth's rotation [10].

PO It is one type of satellites in which a satellite passes above or nearly above both poles of the body being orbited (usually a planet such as the Earth [11].

As the company progressed, other orbits were introduced as launch targets



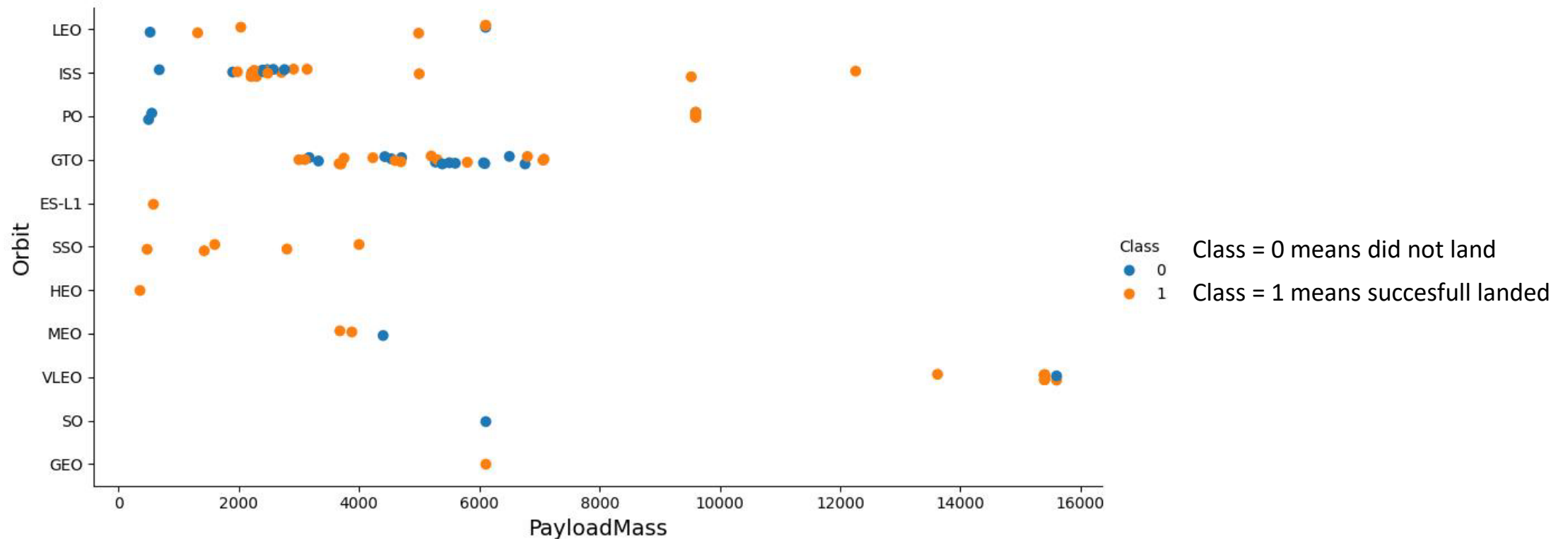
The weighted average of the different success rates associated with these orbits results in 73%



Twenty-seven launches to geostationary transfer orbit (GTO) were not considered because it is a transfer orbit and not a geostationary orbit in itself.

GTO(Geostationary Transfer Orbit): A geostationary transfer orbit is an elliptical Earth orbit used to transfer satellites from low Earth orbit (LEO) to geostationary orbit (GEO). In a GTO, the perigee (closest point to Earth) is much lower than GEO altitude, while the apogee (farthest point) reaches approximately 22,236 miles (35,786 kilometers) above Earth's equator — the altitude of a geostationary orbit. Satellites in GTO use onboard propulsion to circularize their orbit at GEO altitude, where they can provide services such as weather monitoring, communications, and surveillance. [3] .

The successful or positive landing rate is higher for Polar, LEO, and ISS orbits.



For geostationary transfer orbits (GTOs), sought several times during the analysis period, we cannot clearly distinguish between them, as both successful and unsuccessful landings (unsuccessful missions) occur in this region.

With subsidies from EDA, we selected the features that will be used in predicting success in the future module

Field	Data Type	Description	Example
FlightNumber	Integer	Sequential number of the launch	1, 2, 3
PayloadMass	Float	Payload mass in kilograms	5496.0
Orbit	String	Target orbit type of the mission	LEO, GTO, ISS
LaunchSite	String	Launch site location	KSC LC-39A
Flights	Integer	Number of previous flights of the booster	1, 2, 3
GridFins	Boolean	Indicates whether grid fins were used (aerodynamic control surfaces)	True / False
Reused	Boolean	Indicates whether the booster was reused	True / False
Legs	Boolean	Indicates whether the booster has landing legs	True / False
LandingPad	String	Landing pad used for recovery	LZ-1, ASDS OCISLY
Block	Float / Integer	Falcon 9 block version	5.0
ReusedCount	Integer	Number of times the booster has been reused	0, 1, 2
Serial	String	Unique booster identifier	B1049

[15]:	FlightNumber	PayloadMass	Orbit	LaunchSite	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial
0	1	6104.959412	LEO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0003
1	2	525.000000	LEO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0005
2	3	677.000000	ISS	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0007
3	4	500.000000	PO	VAFB SLC 4E	1	False	False	False	NaN	1.0	0	B1003

- To build the model, it is necessary to process the categorical variables 'Orbit', 'LaunchSite', 'LandingPad', 'Serial',
- and then convert all the numeric data to float64

Thank you!

